

Rebuilding the SMP Branch from xPack-Development

**Analysis, Architectural Invariants & Step-by-Step
Implementation Runbook**

Rebuilding the SMP Branch from xpack-development

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Rebuilding the SMP Branch from `xpack-development`

Comprehensive Architectural Analysis, Step-by-Step Implementation Runbook & GitHub Pull Request Blueprint

1. Executive Summary & The GitHub PR Paradigm

1.1 The Genesis & Pitfalls of Monolithic SMP Development

When the multi-core Symmetric Multiprocessing (`smp`) branch was originally developed for $\mu\text{OS++}$ IIIe, it was implemented as a monolithic, sweeping conversion on top of `xpack-development`. In that initial push, hundreds of disparate changes were introduced simultaneously across multiple git repositories: 1. **Unrelated Single-Core Defect Fixes** (POSIX `dirent.h` ISO C empty struct, double-list iterator semantics, newlib syscall types, memory allocator usable size, steady clock duration 64-bit overflow) were interleaved with multi-core scheduler commits. 2. **Architectural Restructuring** (dissolution of `micro-os-plus-iii-devices`, migrating SoC headers into architecture ports) was commingled with thread management state machines. 3. **Silicon Extensions** (ARM Cortex-M33, Raspberry Pi RP2350, AArch32, and AArch64 ports) were added in large batches alongside test harness modifications.

While the resulting `smp` branch in `~/Work2/micro-os-plus/` achieved full multi-core functionality and passing tests, its commit history is **difficult to review, bisect, cherry-pick, or upstream via standard GitHub Pull Requests**. Upstream maintainers cannot easily audit a 10,000-line diff spanning 7 repositories to verify that uniprocessor behavior remains pristine.

1.2 The New Paradigm: Small, Focused, Logically Related Commits

This document establishes the **New SMP Development Procedure**. Under this methodology, the `smp` branch is rebuilt from scratch, starting directly from clean upstream `origin/xpack-development` branches across all repositories.

Every step in this rebuilt history adheres strictly to the **GitHub Pull Request paradigm**: - **Single-Subject Focus**: Every commit addresses exactly **one** issue, defect, feature, or architectural refactoring. Unrelated changes are never commingled. - **Independent Reviewability**: Each commit contains its own clear rationale, problem description, failure mode averted, and unified diff that can be reviewed in isolation. - **Bi-directional Propagation**: Commits developed on the new `smp` branch can be cleanly cherry-picked or opened as isolated Pull Requests targeting `xpack-development`. - **Test Framework Discipline**: Every commit compiles cleanly and passes the rigorous $\mu\text{OS++}$ test framework gate (72/72 baseline uniprocessor tests, followed by multi-core suites as capabilities land).

```
+-----+
|                                     |
|                                     |
| THE NEW REBUILD & PROPAGATION PARADIGM |
|                                     |
|                                     |
+-----+
```

```

+-----+
|                                     |
| UPSTREAM REPOS (xpack-development) <-----+
|                                     |
| |                                     | Clean Cherry-Picks &
| | git clone & checkout                | Focused GitHub PRs
| v                                     | (PR #1 .. PR #32)
| REBUILD WORKSPACE ($WORK)           |
|                                     |
| +---> Phase 0: Dissolve `devices` subtree
|                                     |
| +---> Phase 1: 22 Single-Core Bugfixes & Hardening (PR #1..#22) -----+
| |                                     | Gate: 72/72 Uniprocessor tests 100% GREEN
| |                                     |
| +---> Phase 2: Common SMP Scheduler & Host native-smp (PR #23) -----+
| |                                     | Gate: unifdef zero-diff + 72/72 baseline + native-smp (3/3)
| |                                     |
| +---> Phase 3: Semantic Releases & Modern Silicon Ports (PR #24..#27) -----+
| |                                     | Gate: Cortex-M33 (MPS2 AN521 2-core) + RP2350 clean
| |                                     |
| +---> Phase 4: Multi-Core FPU Stress & 4-Core Platforms (PR #28..#29) -----+
| |                                     | Gate: fp-switch + test-smp-all (Native, M33, AArch32/64)
| |                                     |
| +---> Phase 5: Documentation Suite & Hardware Wiring (PR #30..#32) -----+
| |                                     | Final Parity: 100% Identical to ~/Work2/micro-os-plus/
| |                                     |
+-----+

```

Procedure update (Part II, §11–§18): The rebuild is carried out **only with git commits**. No bash or Python scripts are used: no `scripts/smp/`, no chunk recipes, no stashes and no `git checkout golden -- <files>` lifts. Each commit only modifies or adds files, covers one subject (one fix, one feature, one test, one board, one document), and becomes one GitHub pull request. §11–§18 are the authoritative per-repository commit lists for `micro-os-plus-iii`, `cortexm`, `posix-arch`, `aarch32` and `aarch64`, including the retirement of `devices`. Where §3, §9 and §10 disagree with them, §11–§18 apply.

Execution record (Part III, §20–§22): Part II was executed on 2026-10-06. §20 summarizes every step from `xpack-development` (single CPU) to `smp` (SMP) across all repositories. §21 records the deviations from the plan, the builds, the parity check and the final test results. §22 lists every commit of every repository (kernel, posix-arch, cortexm, aarch32, aarch64, devices) as a pull request: number, head and base branch, SHA, category, stage/theme and subject, plus the cross-repository merge order.

1.3 Target Architecture: The Five Clean Repositories & Dissolution of `devices`

In the original architecture, device-specific drivers and SoC registers were segregated into a separate `micro-os-plus-iii-devices` repository. This model introduced unnecessary cross-repository coupling and circular dependencies during multi-core bringup.

In the new architecture-specific model, `micro-os-plus-iii-devices` is **dissolved**. Its SoC and driver contents are permanently folded into their owning architecture repositories:

<code>micro-os-plus-iii/</code>	(Kernel, portable POSIX I/O, core test framework)
<code>micro-os-plus-iii-cortexm/</code>	(ARMv7-M & ARMv8-M Cortex-M architectures + STM32 / RP2350 SoCs)
<code>micro-os-plus-iii-aarch32/</code>	(ARMv7-A 32-bit architecture + BCM2837 / Raspberry Pi SoCs)
<code>micro-os-plus-iii-aarch64/</code>	(ARMv8-A 64-bit architecture + BCM2837 / Raspberry Pi SoCs)
<code>micro-os-plus-iii-posix-arch/</code>	(Host POSIX architecture + native uniprocessor & native-smp ports)

Compatibility shims (`micro-os-plus::devices` CMake ALIAS targets) ensure that existing external projects referencing the legacy target continue to configure and link without modification.

Crucial Architecture Decision: Following the dissolution of its contents into the respective architecture repositories, the `micro-os-plus-iii-devices` Git repository and folder are **permanently deleted and eliminated** from the active workspace. The final ecosystem consists strictly of the five primary architecture and kernel repositories.

2. Repository Architecture & Workspace Bootstrap Procedure

2.1 Workspace Directory Layout

Rebuilding the SMP branch cleanly requires an isolated workspace folder (`$WORK`), distinct from the reference workspace in `~/Work2/micro-os-plus/`.

```
# Recommended workspace setup
export WORK="/home/dan/Work2/micro-os-plus-rebuild"
mkdir -p "$WORK" && cd "$WORK"
```

The workspace contains the five primary active repositories, plus the legacy `devices` repository used strictly during Phase 0:

Directory	Repository Name	Initial Branch	Role in Rebuild
<code>\$WORK/micro-os-plus-iii</code>	<code>micro-os-plus-iii</code>	<code>xpack-development</code>	Kernel, schedulers, POSIX I/O, test suites
<code>\$WORK/micro-os-plus-iii-cortexm</code>	<code>micro-os-plus-iii-cortexm</code>	<code>xpack-development</code>	Cortex-M0/M3/M4/M7 baseline + M33/RP2350 SMP
<code>\$WORK/micro-os-plus-iii-posix-arch</code>	<code>micro-os-plus-iii-posix-arch</code>	<code>xpack-development</code>	Host POSIX uniprocessor + dual-core <code>native-smp</code>
<code>\$WORK/micro-os-plus-iii-aarch32</code>	<code>micro-os-plus-iii-aarch32</code>	<code>xpack-development</code>	ARMv7-A 32-bit Cortex-A multi-core port
<code>\$WORK/micro-os-plus-iii-aarch64</code>	<code>micro-os-plus-iii-aarch64</code>	<code>xpack-development</code>	ARMv8-A 64-bit Cortex-A multi-core port
<code>\$WORK/micro-os-plus-iii-devices</code>	<code>micro-os-plus-iii-devices</code>	<code>smp</code> / <code>master</code>	Temporary source for Phase 0 subtree migration (permanently deleted after dissolution)

2.2 Cloning Repositories & Branch Topology

All repositories are cloned from their canonical remotes. The clean baseline is checked out, and a dedicated rebuild branch (named `smp-rebuild` or `smp`) is created:

```
cd "$WORK"

# 1. Clone the core repositories
git clone https://github.com/micro-os-plus/micro-os-plus-iii.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-cortexm.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-posix-arch.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-aarch32.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-aarch64.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-devices.git
```

```
# 2. Add local reference remotes pointing to golden source in ~/Work2/micro-os-plus/
for repo in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-iii-posix-arch \
    micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
    cd "$WORK/$repo"
    git remote add golden "/home/dan/Work2/micro-os-plus/$repo"
    git fetch golden
    git checkout -b smp origin/xpack-development
done
```

2.3 Toolchain Verification (`check-env.sh`)

Before applying changes, verify that the environment has all required native and cross-compilers, debuggers, QEMU emulators, and build utilities:

```
cd "$WORK/micro-os-plus-iii"
# Execute environment checker
bash scripts/smp/check-env.sh
```

Required toolchain components: - **Build Tools:** CMake `$\ge 3.20$`, Ninja `$\ge 1.10$`, xPacks Package Manager (`xpm`) `$\ge 0.14$`. - **Host Compilers:** GCC 11, 12, 13, or 14; Clang 16, 17, 18, or 19. - **Cross Toolchains:** `arm-none-eabi-gcc` `$\ge 12.3$`, `aarch64-none-elf-gcc` `$\ge 12.3$`. - **Emulators:** `qemu-system-arm` `$\ge 7.0$` (supporting `mps2-an521`, `raspi3b`), `qemu-system-aarch64` `$\ge 7.0$`. - **Utilities:** `unifdef` (for single-core projection verification), Python 3 with `pypdf`, `weasyprint`, and `markdown`.

2.4 Development Linking Protocol (`xpm link`)

In the μ OS++ development workflow, repositories locate each other via `xpm link`. This ensures that local edits in architecture ports are immediately picked up when compiling kernel test suites:

```
# Register port development links
cd "$WORK/micro-os-plus-iii-posix-arch" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-cortexm" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch32" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch64" && xpm install && xpm link

# Link into the kernel test harness
cd "$WORK/micro-os-plus-iii"
xpm link @micro-os-plus/cortexm
xpm link @micro-os-plus/posix-arch
```

2.5 The Test Framework Paradigm of `micro-os-plus-iii`

The test paradigm in μ OS++ IIIe is strictly structured: 1. **The 72-Test Baseline Matrix:** Comprises 24 distinct build configurations (combining native Linux GCC/Clang with Cortex-M0, M3, M4, M7 cross-compilations) each executing three standardized suites: - `rtos-apis-test`: Comprehensive functional validation of all POSIX and RTOS primitives. - `mutex-stress-test`: High-contention lock/unlock concurrency verification. - `cmsis-os-validator-test`: ARM official CMSIS-RTOS validation suite (60/60 test cases). - **Baseline Invariant:** At every *commit* in Phase 1, `xpm run test-all` must produce 72/72 PASS (100% green). 2. **The Multi-Core SMP Test Matrix (`test-smp-all`):** - `test-native-smp`: Real dual-core host execution using pthreads-as-cores and signal-based IPIs. - `test-2xcortex-m33-cmake`: Dual-core ARMv8-M execution on QEMU `mps2-an521`. - `test-aarch32-rpi3b`: 4-core ARMv7-A execution on QEMU `raspi3b`. - `test-aarch64-rpi3b`: 4-core ARMv8-A execution on QEMU `raspi3b`. - `fp-switch`: Floating-point register corruption stress test under aggressive preemption.

3. Phase 0: Pre-Migration Subtree Dissolution of Legacy

`devices`

3.1 Why the `devices` Repository Must Disappear

The legacy `micro-os-plus-iii-devices` repository isolated silicon device drivers (`soc/stm32f4xx`, `soc/rp2350`, `soc/bcm2837`) outside their corresponding CPU architecture repositories. This caused: - Split architectural headers: Cortex-M33 register definitions lived in `cortexm`, but RP2350 SIO spinlocks lived in `devices`. - Multi-repository synchronization locks: Modifying an interrupt controller required coordinated commits across two repositories before tests could build. - Divergence from upstream xPack distribution packaging.

By folding device support directly into `cortexm`, `posix-arch`, `aarch32`, and `aarch64`, each architecture repository becomes a complete, self-contained hardware abstraction layer.

3.2 History-Preserving `git subtree` Extraction Protocol

Dissolution must **not** be performed via flat file copying; commit history must be 100% preserved. This is achieved using `git subtree add`:

```
cd "$WORK/micro-os-plus-iii-cortexm"

# Subtree add Cortex-M silicon devices from devices repo
git subtree add --prefix=devices "$WORK/micro-os-plus-iii-devices" master

# Restructure into standard architecture directories
mv devices/soc/stm32f4xx soc/stm32f4xx
mv devices/soc/rp2350 soc/rp2350
mv devices/drivers drivers
rm -rf devices

git add soc drivers
git commit -m "refactor(devices): fold Cortex-M SoC and drivers subtree into cortexm port"
```

The identical procedure is executed for `posix-arch` (folding `soc/native`), `aarch32` (folding `soc/bcm2837`), and `aarch64`.

3.3 CMake Target Realignment & Removal of `UOS_DEVICES_DIR`

Prior to dissolution, both `posix-arch` and `cortexm` expected an external sibling `micro-os-plus-iii-devices` directory and contained the following legacy boilerplate in their root `CMakeLists.txt`:

```

set (UOS_DEVICES_DIR "${uos_siblings}/micro-os-plus-iii-devices"
    CACHE PATH "µOS++ III devices working copy")

foreach (_dep_var IN ITEMS UOS_SMP_DIR UOS_DEVICES_DIR)
    ...
    if (NOT EXISTS "${${_dep_var}}/CMakeLists.txt")
        message (FATAL_ERROR "Cannot find the ${_repo} repository at ${${_dep_var}}...")
    endif ()
endforeach ()

add_subdirectory ("${UOS_DEVICES_DIR}" micro-os-plus-iii-devices)

```

If left unchanged after `micro-os-plus-iii-devices` is deleted, any subsequent CMake configure or test command fails immediately with:

```

CMake Error at /tmp/micro-os-plus-iii-posix-arch/CMakeLists.txt:

  Cannot find the devices repository at /tmp/micro-os-plus-iii-devices.

```

Realignment in `micro-os-plus-iii-posix-arch/CMakeLists.txt`

1. Remove `UOS_DEVICES_DIR` and its validation loop from the dependency checks.
2. Define the local block device and filesystem targets directly using the absorbed `drivers/` and `soc/native/` files:

```

# Dissolved devices support: hostfile block device and flatfs
if (NOT TARGET micro-os-plus::devices-hostfile)
    add_library (micro-os-plus-iii-devices-hostfile-interface INTERFACE)
    target_include_directories (
        micro-os-plus-iii-devices-hostfile-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/native/include"
    )
    target_compile_definitions (
        micro-os-plus-iii-devices-hostfile-interface INTERFACE SD_BACKEND_HOSTFILE
    )
    target_sources (
        micro-os-plus-iii-devices-hostfile-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/native/src/sd_hostfile.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
    )
    add_library (micro-os-plus::devices-hostfile ALIAS
        micro-os-plus-iii-devices-hostfile-interface)
endif ()

```

```

# Compatibility alias for full devices suite (FatFs + block device)
if (NOT TARGET micro-os-plus::devices)
    add_library (micro-os-plus-iii-devices-interface INTERFACE)
    target_include_directories (
        micro-os-plus-iii-devices-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs"
    )
    target_sources (
        micro-os-plus-iii-devices-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/sd.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/usb_dwc2.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/ff.c"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/diskio_sd.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/fatfs_hw.cpp"
    )
    add_library (micro-os-plus::devices ALIAS micro-os-plus-iii-devices-interface)
endif ()

```

Realignment in `micro-os-plus-iii-cortexm/CMakeLists.txt`

1. Remove `UOS_DEVICES_DIR` and its validation loop from the dependency checks.
2. Define the local STM32F4xx and RP2350 silicon support targets directly using the absorbed `soc/` files:

```

# Dissolved SoC targets: STM32F4xx and RP2350
if (NOT TARGET micro-os-plus::soc-stm32f411xe)
    add_library (micro-os-plus-iii-soc-stm32f4xx-common-interface INTERFACE)
    target_include_directories (
        micro-os-plus-iii-soc-stm32f4xx-common-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/include"
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/include/cmsis"
    )
    target_sources (
        micro-os-plus-iii-soc-stm32f4xx-common-interface
        INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/system_stm32f4xx.c"
    )

    add_library (micro-os-plus-iii-soc-stm32f411xe-interface INTERFACE)
    target_link_libraries (

```

```

    micro-os-plus-iii-soc-stm32f411xe-interface
    INTERFACE micro-os-plus-iii-soc-stm32f4xx-common-interface
)
target_sources (
    micro-os-plus-iii-soc-stm32f411xe-interface
    INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/vectors_stm32f411xe.c"
)
add_library (micro-os-plus::soc-stm32f411xe
    ALIAS micro-os-plus-iii-soc-stm32f411xe-interface)

add_library (micro-os-plus-iii-soc-stm32f412rx-interface INTERFACE)
target_link_libraries (
    micro-os-plus-iii-soc-stm32f412rx-interface
    INTERFACE micro-os-plus-iii-soc-stm32f4xx-common-interface
)
target_sources (
    micro-os-plus-iii-soc-stm32f412rx-interface
    INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/vectors_stm32f412rx.c"
)
add_library (micro-os-plus::soc-stm32f412rx
    ALIAS micro-os-plus-iii-soc-stm32f412rx-interface)
endif ()

if (NOT TARGET micro-os-plus::soc-rp2350)
    add_library (micro-os-plus-iii-soc-rp2350-interface INTERFACE)
    target_include_directories (
        micro-os-plus-iii-soc-rp2350-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/include"
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/include/cmsis"
    )
    target_sources (
        micro-os-plus-iii-soc-rp2350-interface
        INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/src/system_rp2350.c"
    )
    add_library (micro-os-plus::soc-rp2350
        ALIAS micro-os-plus-iii-soc-rp2350-interface)
endif ()

```

Realignment in `aarch32` and `aarch64`

Both `micro-os-plus-iii-aarch32/CMakeLists.txt` and `micro-os-plus-iii-aarch64/CMakeLists.txt` compile the dissolved `soc/bcm2837` sources locally into `micro-os-plus::aarch32` and `micro-os-plus::aarch64` interface targets, with zero external dependency on `devices`.

Furthermore, following the dissolution of `micro-os-plus-iii-devices`, tests exercising the SD host (`sd_test`, `smp-mat-sdcard-test`), USB controller (`usb_test`), and file systems require the `micro-os-plus::devices` target. Both AArch32 and AArch64 declare this target locally from their absorbed `drivers/` and `soc/` trees:

```
# Dissolved devices support: SD host, USB, and FatFs stack
if (NOT TARGET micro-os-plus::devices)
    add_library (micro-os-plus-iii-devices-interface INTERFACE)
    target_include_directories (
        micro-os-plus-iii-devices-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs"
    )
    target_sources (
        micro-os-plus-iii-devices-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/sd.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/usb_dwc2.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/ff.c"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/diskio_sd.cpp"
            "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/fatfs_hw.cpp"
    )
    add_library (micro-os-plus::devices ALIAS micro-os-plus-iii-devices-interface)
endif ()
```

Duplicate Port Target Prevention in Platform Harness

In the μ OS++ test framework (`tests/cmake/tests-main.cmake`), the architecture port under test is added directly via `add_subdirectory ("${UOS_<PORT>_DIR}" "port-bin")`. Therefore: - The architecture port (`micro-os-plus-iii-aarch32` or `micro-os-plus-iii-aarch64`) must **never** be listed in `dependencies-folders.cmake` (`xpack_dependencies_folders`). - Listing the port in `dependencies-folders.cmake` causes CMake policy `CMP0002` violations (`add_library cannot create target ... because another target with the same name already exists`). - Platforms list only test suite directories and portable xPacks (`mutex-stress`, `rtos-apis`, `arm-cmsis-rtos-validator`, `chan-fatfs`) in `dependencies-folders.cmake`.

Shared Test Support Scaffolding Realignment (`test_smpl` $\rightarrow$ `tests/smp-support`)

When the shared test scaffolding was consolidated into `tests/smp-support/` (`hw_result.hpp` and `board-contract.cpp`): - `micro-os-plus::test-support` in `micro-os-plus-iii/CMakeLists.txt` exposes `${CMAKE_CURRENT_SOURCE_DIR}/tests/smp-support/include` (with a fallback to `test_smpl/include`). - A backwards-compatibility symlink `test_smpl -> tests/smp-support` is maintained in the

root of `micro-os-plus-iii` so that legacy test runners (`run-hw.sh`, `run-host.sh`, `run-qemu.sh`) and scripts referencing `$SMP_DIR/test_smpl/` execute without error.

3.4 Phase 0 Verification Gate

```
# Verify all existing uniprocessor boards configure and build cleanly
cd "$WORK/micro-os-plus-iii"
xpm run test-all # Must remain 72/72 PASS
```

3.5 Permanent Deletion of the Legacy `devices` Repository and Working Directory

Once the subtree extraction into `cortexm`, `posix-arch`, `aarch32`, and `aarch64` is complete and verified, the `micro-os-plus-iii-devices` repository and working directory have fulfilled their transitional purpose.

The `micro-os-plus-iii-devices` Git clone/directory must be **permanently deleted from the active workspace**:

```
# Permanently delete the dissolved devices repository clone from the workspace
rm -rf "$WORK/micro-os-plus-iii-devices"
```

Key Architectural Rules Post-Deletion:

- 1. Zero External Device Dependency:** No architecture repository or kernel build configuration may reference or depend on an external `micro-os-plus-iii-devices` directory.
- 2. Self-Contained Ports:** Silicon drivers (`soc/*`) and neutral drivers (`drivers/*`) are compiled and maintained directly inside each architecture repository (`cortexm`, `posix-arch`, `aarch32`, `aarch64`).
- 3. Retired Repository:** The standalone `micro-os-plus-iii-devices` repository is officially retired and archived; it is never propagated into future `xpack-development` or `smp` deployment cycles.

4. Phase 1: Single-Core Defect Corrections & Hardening (PR #1 through PR #22)

Phase 1 isolates all bugfixes, standard conformance corrections, and allocator hardening that are **pure uniprocessor improvements**. Landing these 22 commits first guarantees that single-core $\mu\text{OS}++$ is completely defect-free before any multi-core scheduling logic is introduced.

Summary Table of Phase 1 Pull Requests

PR #	Subject / Theme	File(s) Touched	Problem & Failure Mode Averted	Verification Gate
#1	POSIX <code>dirent.h</code> ISO C Conformance	<code>include/cmsis-plus/posix/dirent.h</code>	Empty struct <code>typedef struct { ; } DIR;</code> violates ISO C99/C11 §6.7.2.1; triggers <code>-Wextra-semi</code> .	72/72 pass; clean <code>-Wextra-semi</code>
#2	List Iterator Access Semantics	<code>include/cmsis-plus/utils/lists.h</code>	<code>node_->next</code> accessed as data member instead of method <code>node_->next()</code> , breaking C++17/20 concepts.	72/72 pass; template concepts clean
#3	Newlib Syscall Return Types	<code>posix-io/c-syscalls-aliases-standard.h</code>	Newlib 64-bit uses <code>_READ_WRITE_RETURN_TYPE (int or _ssize_t)</code> , causing conflicting weak alias signatures.	72/72 pass; reentrant tests clean
#4	Export Thread Suspend Symbol	<code>src/rtos/os-thread.cpp</code>	Out-of-line <code>inline void suspend()</code> prevented symbol emission; undefined reference in C API wrapper.	72/72 pass; wrapper links cleanly
#5	Memory Allocator Usable Size	<code>os-memory.*</code> , <code>malloc.cpp</code> , <code>first-fit-top.*</code>	Missing <code>do_usable_size()</code> broke <code>realloc()</code> ; <code>calloc()</code> lacked multiplication overflow checks.	72/72 pass; usable size verified
#6	C++17 Aligned Allocation	<code>src/libcpp/new.cpp</code>	Missing standard <code>operator new/delete(align_val_t)</code> overloads for over-aligned types.	72/72 pass; aligned new clean
#7	Static Error Category Storage	<code>src/libcpp/system-error.cpp</code>	Temporary category objects left dangling pointers inside <code>throw std::system_error</code> .	72/72 pass; static lifetime clean
#8	Chrono 64-Bit Duration Overflow	<code>src/libcpp/chrono.cpp</code>	<code>cycles * 1e9 / freq</code> overflowed 64-bit int after ~13 minutes. Decompose into seconds and remainder.	72/72 pass; chrono math overflow-free
#9	CMSIS-RTOS C Wrapper Cleanups	<code>src/rtos/os-c-wrapper.cpp</code>	Default timer must be one-shot; polymorphic mutex delete; cast <code>millisec</code> to	72/72 pass; validator 60/60 clean

PR #	Subject / Theme	File(s) Touched	Problem & Failure Mode Averted	Verification Gate
			<code>uint64_t</code> prior to <code>*</code> <code>1000</code> .	
#10	POSIX I/O FD Table Mutexing	<code>src/posix-io/file-descriptors-manager.cpp</code>	Unprotected descriptor table corrupted under concurrent multi-threaded <code>open()</code> / <code>close()</code> .	72/72 pass; concurrent I/O stress pass
#11	POSIX I/O Deferred List Locking	<code>posix-io/file-system.h</code> , <code>net-stack.h</code>	Concurrent recycling of deferred file/socket nodes caused corrupted intrusive list links.	72/72 pass; list recycling clean
#12	Block Device Sector Size Check	<code>src/posix-io/block-device.cpp</code>	Inverted check in <code>vfcntl</code> : tested <code>!= 0</code> instead of <code>== 0</code> , falsely rejecting valid block devices.	72/72 pass; valid block size returned
#13	ARMv8-M Semihosting Traps	<code>include/cmsis-plus/arm/semihosting.h</code>	Missing <code>__ARM_ARCH_8M_MAIN</code> macro guards and <code>SEMIHOST_TRAP_HLT</code> support.	72/72 pass; ARM semihosting pass
#14	ARMv8-M Exception Handlers	<code>src/startup/exception-handlers.c</code>	SCB register setup and fault dump routines lacked <code>__ARM_ARCH_8M_MAIN</code> guards.	72/72 pass; fault handlers compile
#15	Semihosting <code>fstat()</code> & Mirror	<code>src/semihosting/c-syscalls-semihosting.cpp</code>	Assign <code>S_IFCHR</code> only when mode unset; add weak <code>os_board_console_merror()</code> hook.	72/72 pass; console mirror clean
#16	Timer Callback Decoupling	<code>internal/os-lists.cpp</code> , <code>os-timer.cpp</code>	Executing timer callbacks inside critical section caused deadlocks; drift catch-up guard.	72/72 pass; timer callback mutexes clean
#17	Mutex Priority Ceiling Protocol	<code>src/rtos/os-mutex.cpp</code>	Enforce ceiling before acquiring ownership; track maximum boost among multiple waiters.	72/72 pass; <code>mutex-stress-test</code> pass
#18	Thread Lifecycle State & Reaper	<code>os-c-decls.h</code> , <code>os-thread.h</code> , <code>os-idle.cpp</code>	Add <code>state::destroying = 7</code> ; idle thread reaper safely validates no core is running thread before free.	72/72 pass; thread reaper green
#19			Atomic link node & mutex unlock under	

PR #	Subject / Theme	File(s) Touched	Problem & Failure Mode Averted	Verification Gate
	CondVar Lost-Signal Elimination	<code>src/rtos/os-condvar.cpp</code> , <code>os-condvar.h</code>	<code>scheduler::critical_section</code> ; bind condvar to clock.	72/72 pass; condvar concurrency green
#20	<code>std::thread</code> Handle Sync	<code>thread-cpp.h</code> , <code>thread_internal.h</code>	<code>thread::join()</code> waits on native thread handle; retain functor object to eliminate use-after-free.	72/72 pass; functor lifetime verified
#21	MQueue Preemptive Reschedule	<code>src/rtos/os-mqueue.cpp</code>	Call <code>port::scheduler::reschedule()</code> immediately after send/receive wakeups.	72/72 pass; queue preemption green
#22	High-Res Clock Port Sync	<code>os-decls.h</code> , <code>os-clocks.cpp</code> , Port inlines	Declares <code>has_hardware_counter()</code> ; fast-paths <code>now()</code> ; SysTick ICSR pending overflow inspection.	72/72 pass; ICSR pending fix verified

5. Phase 2: Common Multi-Core SMP Kernel Infrastructure & Host Runtime (PR #23 / Step 14)

5.1 Architectural Foundations of the SMP Kernel

Phase 2 constitutes the architectural core of the entire SMP migration. It transitions the kernel from a single-CPU RTOS to a Symmetric Multiprocessing engine capable of executing threads concurrently across N physical cores.

Key data structure transformations in `include/cmsis-plus/rtos/os-sched.h` and `src/rtos/os-core.cpp`:

- **Per-CPU Current Threads**: Replaces scalar `current_thread_` with an array indexed by `port_cpu_id()`:
`cpp #if defined(OS_USE_SMP_SCHEDULER) extern thread* volatile current_thread_[OS_NCPU]; #else extern thread* volatile current_thread_; #endif`
- **Per-CPU Ready Lists**: Replaces the monolithic ready list with `ready_list_[OS_NCPU]`, eliminating global queue lock bottlenecking during normal thread execution.
- **Recursive Kernel Spinlock** (`_smp_klock`): Protects scheduler state transitions. Backed by architecture hardware spinlocks (e.g. RP2350 SIO spinlock 0, Cortex-M33 LDREX/STREX, or POSIX mutex).

5.2 The 5-Stage Stack Pointer Claim/Publish Protocol

To prevent fatal stack corruption when two cores attempt to switch to or from the same thread simultaneously, the scheduler implements a strict 5-stage handshake:

1. **Claim**: The incoming thread stack pointer is atomically set to `nullptr` on the target core, marking it actively running.
2. **Spill**: The outgoing thread's callee-saved registers are pushed to its current stack.
3. **Switch**: Hardware Stack Pointer register (SP) is updated to the incoming thread's stack.
4. **Publish**: The outgoing thread's saved stack pointer is published to `from->context_.port_.stack_ptr = sp` **only after** state is completely saved.
5. **Restore**: Callee-saved registers are popped from the new stack, and thread execution resumes.

5.3 Prohibiting Context Switch While Holding Spinlock

A primary invariant in $\mu\text{OS++}$ SMP: **A hardware context switch must never be executed while holding `_smp_klock`**. Context switching while holding the kernel spinlock causes deadlocks across cores. Instead, `internal_switch_threads()` schedules the switch, marks the CPU as requiring rescheduling, releases `_smp_klock`, and allows the architecture PendSV interrupt or signal handler to execute the actual register swap.

5.4 Phase 2 Verification Gates

1. The `unifdef` **Single-Core Invariant**: Running `unifdef -UOS_USE_SMP_SCHEDULER` on the Phase 2 kernel source must produce a tree that is **100% byte-identical to Step 22**. Uniprocessor builds incur zero code bloat and zero overhead.
2. **72/72 Uniprocessor Test Gate**: `xpm run test-all` passes with 100% green status.

3. **Genuine Dual-Core Host Verification (`native-smp`):** Executes `tests/platforms/native-smp/` under Linux host using 2 OS threads as virtual CPU cores: `bash cd "$WORK/micro-os-plus-iii/tests" xpm run test-native-smp # Must execute and pass: # 1. rtos-apis-test (dual-core concurrency) # 2. mutex-stress-test (contention across cores) # 3. cmsis-os-validator-test (60/60 test cases)`

6. Phase 3: Semantic Versioning, Port Releases & Silicon Extensions (PR #24 through PR #27)

6.1 PR #24: Port Package Semantic Releases

Publishes the formal multi-core hardware abstraction contract: - `micro-os-plus-iii-posix-arch`: Version bumped from `1.0.1` $\rightarrow$ `1.1.0`. - `micro-os-plus-iii-cortexm`: Version bumped from `1.1.0` $\rightarrow$ `1.2.0`. Local semantic git tags (`v1.1.0`, `v1.2.0`) are generated for repeatable downstream dependency resolution.

6.2 PR #25: Execution of Devices Subtree Dissolution, CMake Realignment & Repository Deletion

Finalizes the historical merge of `soc/*` and `drivers/*` into the owning architecture ports, establishing clean build targets and permanent ALIAS compatibility shims.

Key Implementation Actions:

- Subtree Merge:** Folds `soc/stm32f4xx` and `soc/rp2350` into `cortexm`, `soc/native` and `drivers/*` into `posix-arch`, and `soc/bcm2837` into `aarch32` / `aarch64`.
- CMakeLists.txt Realignment:** - In `posix-arch/CMakeLists.txt`: Remove `UOS_DEVICES_DIR` dependency check. Declare `micro-os-plus::devices-hostfile` (from `soc/native/src/sd_hostfile.cpp` and `drivers/src/flatfs.cpp` with `SD_BACKEND_HOSTFILE`) and `micro-os-plus::devices` directly. - In `cortexm/CMakeLists.txt`: Remove `UOS_DEVICES_DIR` dependency check. Declare `micro-os-plus::soc-stm32f411xe`, `micro-os-plus::soc-stm32f412rx`, and `micro-os-plus::soc-rp2350` directly from the local `soc/` paths.
- Repository Deletion:** Immediately upon completion of the subtree graft and CMake realignment, the `micro-os-plus-iii-devices` repository folder and Git repository are **permanently deleted from the workspace**, retiring the standalone repository and guaranteeing that all downstream projects build and test 100% self-contained.

6.3 PR #26: Modern Multi-Core Silicon Ports (Cortex-M33 & RP2350)

Introduces genuine silicon hardware multi-core implementations: - **ARM Cortex-M33 Port:** Supports ARMv8-M architecture, Message Handling Unit (MHU) Inter-Processor Interrupts (IPI), and per-core SysTick clocks. - **Raspberry Pi RP2350 Port:** Implements recursive `_smp_klock` backed by SIO Hardware Spinlock 0 (`*(volatile uint32_t*)(0xD0000100)`), Core 1 wake-up mailbox, and FIFO IRQ 25. - **Kernel IPI Dispatching:** In `src/rtos/os-thread.cpp`, dispatches wake-up IPIs whenever a thread unblocks on an alternate core (`OS_INTEGER_RTOS_PORT_NCPU > 1`).

6.4 PR #27: Modular Add-Only CMake Build Architecture

Splits the monolithic CMake interface into clean, granular sub-targets: - `micro-os-plus::iii-core`: Bare kernel RTOS engine without POSIX I/O. - `micro-os-plus::iii-posix-io`: File system, mount tables, and character devices. - `micro-os-plus::port-smp-decls`: Pure SMP forward declarations for thin ports. - Adds cross-compilation toolchain profiles (`cmake/toolchains/arm-none-eabi.cmake`, `cmake/`

```
toolchains/aarch64-none-elf.cmake ). - Adds QEMU 2-core platform
tests/platforms/2xcortex-m33/ ( mps2-an521 --cpu cortex-m33 --smp 2 ).
```

7. Phase 4: Multi-Core Test Suites, Multi-Arch Platforms & CI Harness (PR #28 & PR #29)

7.1 PR #28: Multi-Core FPU Stress Suite & Scaffolding Relocation

- **FPU Context Switch Stress Suite** (`tests/sources/fp-switch/`): 6 threads computing mathematical floating-point series across cores under aggressive 1ms preemptive slicing (3000 ticks). Detects missing hardware FPU lazy-stacking or register corruptions.
- **Scaffolding Relocation**: Moves legacy root `test_smp/` into `tests/smp-support/`, cleanly exported as `micro-os-plus::test-support` providing standardized test reporting and hardware exit registers.

7.2 PR #29: Multi-Architecture 4-Core Platforms & Composite CI Action

- **AArch32 & AArch64 4-Core Platforms**: Implements QEMU `raspi3b` platforms (`tests/platforms/aarch32-rpi3b/`, `tests/platforms/aarch64-rpi3b/`) executing across 4 ARM Cortex-A53 cores.
 - **Composite Test Action** (`test-smp-all`): Registered in `tests/package.json`, executing the complete multi-core test matrix in one command:

```
json "test-smp-all": { "description": "Run all SMP test suites across Native, Cortex-M33, and AArch32/64", "actions": [ "xpm run test-native-smp", "xpm run test-2xcortex-m33-cmake", "xpm run test-aarch32-rpi3b", "xpm run test-aarch64-rpi3b" ] }
```
-

8. Phase 5: Documentation Suite, Master Reconciliation & Silicon Wiring (PR #30 through PR #32)

8.1 PR #30: Comprehensive Documentation & Specification Runbooks Suite

Integrates all architectural runbooks and migration specifications into `docs/render-pdfs.sh`. Builds the complete 16-document PDF publication suite.

8.2 PR #31: Final Master Reconciliation & Repository Cleansing

- Merges upstream `origin/xpack-development` cleanly.
- Restores `.github/workflows/ci.yml`, `README.md`, `LICENSE`, and Doxygen templates to be **100% byte-identical** to upstream baseline.
- Purges temporary transition scripts (`scripts/smp/`).

8.3 PR #32: Dedicated Raspberry Pi Pico 2 Physical Hardware Platform Wiring

Adds standalone hardware platform `tests/platforms/cortexm-pico2/` enabling direct OpenOCD SWD flashing and UART verification on physical RP2350 silicon. Implements Core 1 secondary boot handshake via SIO mailbox registers (`sio_hw->fifo_wr` / `sio_hw->fifo_rd`).

9. Operator's Execution Manual: The Step-by-Step Lifting Loop

Superseded. The script-driven loop below (`new-step.sh` , `chunks/stepNN.sh` , `verify-step.sh` , `advance-step.sh`) is kept only as a historical record. The authoritative procedure is the commit-only cycle in §12. Experience with this loop showed that the recipes mix subjects: a `git checkout golden/smp -- <files>` lift staged the files of PR #2 and PR #3 into the commit for PR #1. The scripts also leave transition tooling in the repository, which then has to be removed again (PR #31).

9.1 The Golden 4-Command Development Cycle per Step

For each of the 32 pull requests, the developer executes a standardized four-command loop:

+-----+	+-----+	+-----+	+-----+
1. NEW STEP	---> 2. APPLY CHUNK	---> 3. VERIFY GATE	---> 4. COMMIT / ADV
Fork step branch	Lift clean diff	Compile & Test	Atomic git commit
+-----+	+-----+	+-----+	+-----+

Step 1: Fork the Step Branch

```
cd "$WORK/micro-os-plus-iii"
# Creates step/NN branched from step/NN-1 (or xpack-development at Step 1)
bash scripts/smp/new-step.sh <NN>
```

Step 2: Apply the Step's Isolated Chunk

```
# Sourced directly from golden reference repo in ~/Work2/micro-os-plus/
bash scripts/smp/chunks/step<NN>.sh
# Or inspect unified diff:
bash scripts/smp/show-chunk.sh <NN> <file...>
```

Step 3: Execute the Verification Gate

```
# Validates pristine rules, unifdef projection, and runs 72/72 test matrix
bash scripts/smp/verify-step.sh <NN>
```

Step 4: Commit and Advance

```
# Creates atomic git commit with conventional commit header
bash scripts/smp/advance-step.sh <NN> "commit message description"
```

9.2 Stacking Pull Requests on GitHub

When submitting the 32 steps to GitHub: 1. Push each local step branch: `git push -u origin step/<NN>`. 2. Open Pull Request on GitHub: - For PR #1: Target base `xpack-development` $\rightarrow$ Head `step/01`. - For PR #2: Target base `step/01` $\rightarrow$ Head `step/02`. - For PR #N: Target base `step/<N-1>` $\rightarrow$ Head `step/<N>`. 3. When upstream merges PR #1 into `xpack-development`, use GitHub's web UI to retarget PR #2's base from `step/01` to `xpack-development`.

9.3 Cherry-Picking Commits Back into `xpack-development`

Because each commit is strictly self-contained and touches only files relevant to its single theme, any individual commit or group of commits can be cherry-picked directly into `xpack-development`:

```
cd "$WORK/micro-os-plus-iii"
git checkout xpack-development
git fetch origin

# Example: Cherry-pick PR #1 (dirent) and PR #2 (lists) directly
git cherry-pick <PR_1_COMMIT_HASH>
git cherry-pick <PR_2_COMMIT_HASH>

# Verify baseline tests remain 100% green
xpm run test-all
```

10. Comprehensive Verification Matrix & Acceptance Gates

Every pull request satisfies an explicit, automated acceptance gate prior to being advanced:

PR ID	Theme & Domain	Repositories	Self-Contained?	Automated Gate Command	Acceptance Pass Criteria
PR #1	POSIX <code>dirent.h</code> Conformance	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; <code>-Wextra-semi</code> clean
PR #2	List Iterator Member Semantics	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; template concepts clean
PR #3	Newlib Syscall Return Types	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; newlib reentrant tests pass
PR #4	Thread Suspend Symbol Export	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; C wrapper resolves symbol
PR #5	Memory Allocator Usable Size	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; <code>malloc_usable_size</code> verified
PR #6	C++17 Aligned Allocation	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; aligned <code>operator new</code> clean
PR #7	Static Error Category Storage	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; static storage eliminates dangling ptrs
PR #8	Steady Clock Duration Overflow	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; chrono math overflow-free
PR #9	CMSIS-RTOS C Wrapper Cleanups	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; validator (60/60) clean
PR #10	POSIX I/O FD Table Mutex	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; concurrent I/O stress green
PR #11	POSIX I/O Deferred List Locking	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; list recycling thread-safe
PR #12	Block Device Sector Size Check	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; valid block size returned
PR #13	ARMv8-M Semihosting Traps	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; ARM semihosting tests pass
PR #14	ARMv8-M Exception Handlers	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; ARMv8-M fault vectors compile
PR #15	Semihosting <code>fstat()</code> & Mirror	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; mode preserved &

PR ID	Theme & Domain	Repositories	Self-Contained?	Automated Gate Command	Acceptance Pass Criteria
					console mirror hook
PR #16	Software Timer Decoupling	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; timer callback mutexes clean
PR #17	Mutex Priority Ceiling	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; <code>mutex-stress-test</code> passes
PR #18	Thread Lifecycle & Reaper	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; dynamic thread reaper green
PR #19	CondVar Lost-Signal Elimination	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; condvar concurrency stress green
PR #20	<code>std::thread</code> Handle Sync	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; functor closure lifetime verified
PR #21	MQueue Preemptive Reschedule	Kernel	standalone	<code>xpm run test-all</code>	72/72 tests pass; message queue preemption green
PR #22	High-Res Clock Port Sync	K + C + P	atomic	<code>xpm run test-all</code>	72/72 tests pass; ICSR pending fix verified
PR #23	Multi-Core SMP Core & Host	K + C + P	atomic	<code>scripts/smp/verify-step.sh 14</code>	<code>unifdef</code> zero-diff + 72/72 + <code>native-smp</code> (3/3) PASS
PR #24	Port Semantic Releases	P + C	standalone	<code>git tag -l</code>	<code>v1.1.0</code> and <code>v1.2.0</code> local tags generated
PR #25	Devices Subtree Dissolution	P + C + Arch	standalone	<code>xpm run build ...</code>	Subtree merged; CMake ALIAS targets resolve; <code>devices</code> folder deleted from workspace
PR #26	Modern Silicon Ports	C + K	add-only	<code>scripts/smp/verify-step.sh 25</code>	M33/RP2350 port targets compile cleanly
PR #27	Modular Add-Only CMake	K + C	add-only	<code>scripts/smp/verify-step.sh 26</code>	72/72 baseline + <code>2xcortex-m33</code> (4/4) PASS
PR #28	FPU Context Switch Stress	Kernel	add-only	<code>scripts/smp/verify-step.sh 27</code>	<code>fp-switch</code> passes on <code>2xcortex-m33</code> (4/4)

PR ID	Theme & Domain	Repositories	Self-Contained?	Automated Gate Command	Acceptance Pass Criteria
PR #29	Multi-Arch 4-Core Platforms	K + A32 + A64	add-only	<code>scripts/smp/verify-step.sh 28</code>	<code>test-smp-all</code> passes (Native, M33, AArch)
PR #30	Documentation Suite	Kernel	standalone	<code>docs/render-pdfs.sh</code>	All PDFs compile cleanly without warnings
PR #31	Final Merge & Cleansing	All Repos	clean-merge	<code>bash scripts/smp/finalize.sh</code>	Upstream CI metadata 100% byte-identical
PR #32	Pico 2 Hardware Platform	K + C	add-only	<code>xpm run build --config cortexm-pico2</code>	Pico 2 platform compiles for hardware SWD

10.2 Final Parity Verification with Golden Source

At the conclusion of Step 32: - The rebuilt `smp` branch in `$WORK/micro-os-plus-iii` is compared against `/home/dan/Work2/micro-os-plus/micro-os-plus-iii`: `bash diff -ruN -x ".git" "$WORK/micro-os-plus-iii" "/home/dan/Work2/micro-os-plus/micro-os-plus-iii"` - **Acceptance Invariant:** The source code trees across all five repositories (`micro-os-plus-iii`, `cortexm`, `posix-arch`, `aarch32`, `aarch64`) achieve **100% functional and structural parity** with the reference implementation in `~/Work2/micro-os-plus/`, but possess a **clean, granular, 32-commit GitHub Pull Request history** ready for immediate upstream integration.

Part II — Commit-Only Migration Procedure (Authoritative)

11. Principles: Only Commits, One Subject per Commit, No Scripts

11.1 Why Part I's script loop is replaced

Part I (§9) drives the rebuild with `scripts/smp/*.sh` and `chunks/stepNN.sh` recipes. A dry run on clean clones of `github.com/dan-maio/*` showed three structural problems with it:

1. **Recipes mix subjects.** A recipe lifts whole files with `git checkout golden/smp -- <files>`. When that happens, the lift *stages* every lifted file. Step 01's recipe staged `lists.h` (PR #2) and `c-syscalls-aliases-standard.h` (PR #3) into the commit for PR #1, so the commit no longer matched its PR.
2. **The tooling becomes part of the history.** `scripts/smp/`, its chunks and their `__pycache__` are committed, and later have to be removed again (PR #31). A GitHub reviewer sees tooling churn that has nothing to do with the kernel.
3. **A step's result is not reviewable before it exists.** What lands is whatever the script produced at run time. A reviewer cannot read a recipe and know the exact resulting diff.

Decision: the rebuild uses **git commits only**. Every commit is written as the pull request it will become: one subject, a self-contained message, and a diff that is reviewable on its own. The procedure needs no shell or Python script. The only commands are `git` (including the interactive hunk selection built into git) and `xpm` (the μ OS++ test framework).

11.2 The ten rules

#	Rule	Consequence
R1	One subject per commit: one correction, one feature, one test, one board, one platform, or one document.	A file touched by several subjects is lifted hunk by hunk (<code>git checkout -p</code> , §12.2), never whole.
R2	A commit only modifies or adds files. It never deletes or renames an upstream file.	Golden's deletions (<code>.github/</code> , <code>doxygen/</code> , <code>inspiration/</code> , <code>templates/</code> , <code>.vscode/</code> , <code>.settings/</code> , <code>CHANGELOG.md</code> , <code>README*.md</code> , <code>LICENSE</code> → <code>LICENSE-micro-os-plus-iii</code> , <code>package-lock.json</code> , <code>CLAUDE.md</code> , arch <code>.npmignore</code> / <code>LICENSE</code> / <code>README.md</code> , posix-arch <code>test/trace-*</code>) are not reproduced . PR #31's "restore to upstream" therefore has nothing to do.
R3	No procedure tooling is committed.	<code>scripts/smp/**</code> and <code>tests/xpacks/**</code> (installed packages, created by <code>xpm install</code>) are never committed.
R4	Product files may be shell, procedure files may not.	Test runners that <code>ctest</code> executes (<code>test_smp/run-{host,qemu,hw}.sh</code> , <code>test/boards/*/hw.sh</code> , <code>test/qemu.sh</code> , <code>test/boards/native/run.sh</code>) are product files. They are committed with the test infrastructure that calls them.
R5	A fix to code that this series introduces is folded into the commit that introduces that code.	No PR ships a known bug that a later PR repairs. Fixes to <i>upstream</i> code are separate commits (the Phase 1 corrections).
R6	The provider lands before the consumer.	A port symbol lands before the kernel code that calls it, and kernel declarations land before port code that implements them. When neither order compiles alone, the commits form a group . The gate runs at the group's last commit, and every message in the group names the group (<code>Group: G-hrclock</code>).
R7	Every commit compiles in the single-core configurations.	New SMP code sits under <code>#if defined(OS_USE_SMP_SCHEDULER)</code> . <code>unifdef -UOS_USE_SMP_SCHEDULER</code> of any SMP commit gives the tree of its parent.
R8	New build files grow with their sources.	A commit that adds a <code>.cpp</code> also adds its line to the port's <code>CMakeLists.txt</code> . No commit lists a file that does not exist yet.
R9	Tests are added one per commit.	The port test builders (<code>test/CMakeLists.txt</code>) and the kernel platforms register <i>every directory present</i> under <code>test/<board>/</code> . One new test

#	Rule	Consequence
		directory therefore equals one commit and one new <code>ctest</code> entry.
R10	Every gate is an <code>xpm</code> action of the μ OS++ test framework.	§13.3. No ad-hoc gate script exists.

11.3 Corrections to Part I, checked against the golden `smp` branches

These facts were read from `~/Work2/micro-os-plus/*` (`xpack-development..smp`, read-only):

Part I says	Golden <code>smp</code> actually has	Part II rule
Clone from <code>github.com/micro-os-plus/*</code> (§2.2)	Fork <code>github.com/dan-maio/*</code> . On aarch64, local <code>xpack-development</code> is <code>366c957</code> but GitHub <code>origin/xpack-development</code> is <code>fe904c0</code> .	Clone from <code>dan-maio</code> . Before starting, compare the aarch64 base with <code>git log --oneline 366c957...fe904c0</code> and record which one was used.
<code>xpm link @micro-os-plus/cortexm</code> (§2.4)	The package names are <code>@micro-os-plus/micro-os-plus-iii-{posix-arch,cortexm,aarch32,aarch64}</code> .	Link them per configuration through <code>xpm run install --config <cfg></code> ; each <code>*-actions</code> configuration already runs <code>xpm link</code> . Do not use <code>link-deps-all</code> : it aborts on <code>@micro-os-plus/build-helper</code> .
<code>tests/platforms/native-smp/</code> , <code>test-native-smp</code> (§5.4)	No such platform. <code>tests/platforms/native</code> builds and runs posix-arch's own <code>test/native/*</code> (rtos-apis, mutex-stress, flats, mutex-ceiling, smp-*) through <code>test_smpl/run-host.sh</code> .	Native SMP tests are posix-arch test directories (§16.3). They run under <code>xpm run test-native-cmake</code> .
<code>test-smp-all</code> action (§7.2)	The action is <code>test-smp-cmake</code> , and it is part of <code>test-all</code> (<code>native</code> + <code>cortex</code> + <code>smp</code>).	The first SMP platform commit creates <code>test-smp-cmake</code> and adds it to <code>test-all</code> . Each later platform commit appends one line to it (§15.6).
<code>test_smpl/</code> → <code>tests/smp-support/</code> plus a symlink (§3.3, §7.1)	<code>test_smpl/</code> is a real directory at the kernel root, and the platforms reference <code>../test_smpl</code> .	<code>test_smpl/</code> is added as is (K-H5*). No relocation, no symlink.
Release bumps posix-arch 1.1.0, cortexm 1.2.0 (§6.1)	<code>package.json</code> versions are unchanged: posix-arch 1.0.1, cortexm 1.1.0, aarch32/64 0.1.0.	A version bump is an optional last commit per port (<code>chore(release)</code>), made only when publishing.
PR #25 = devices dissolution (§6.2)	This is the same work as Phase 0.	Done once, as stage S0 (§14).
PR #23 must be one atomic K+C+P commit (§5, files-modif-by-step)	The kernel SMP code is guarded (R7), so single-core builds do not see it.	It is split into themed commits per repository (§15.3, §16.2, §17.2). SMP builds are expected to be green only at the end of stage S2, which is the S2 gate.
Final parity is "100% identical" (§10.2)	Identity is impossible by design: R2/R3 keep upstream files that golden deletes, and the arch repos gain <code>soc/</code> , <code>drivers/</code> that golden only references.	The parity check lists the intended differences (§19.2). Every other difference is a defect.
<code>devices</code> subtree is added under <code>devices/</code> and then moved (§3.2)	The devices repository (<code>github/smp</code> = <code>e8e39d5</code>) has <code>include/</code> , <code>src/</code> , <code>fatfs/</code> and <code>soc/<chip>/</code> at its top level.	It is imported with plain add commits into <code>drivers/include</code> , <code>drivers/src</code> , <code>drivers/fatfs</code> , <code>soc/<chip></code> (§14).

12. The Commit Cycle (Manual, No Scripts)

12.1 Workspace

```
export WORK=/tmp                # or any empty directory
cd "$WORK"

for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-iii-posix-arch \
        micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64 micro-os-plus-iii-devices; do
    git clone https://github.com/dan-maio/$r.git
done

for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-iii-posix-arch \
        micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
    git -C "$WORK/$r" remote add golden "$HOME/Work2/micro-os-plus/$r" # read-only reference
    git -C "$WORK/$r" fetch golden smp                                # fetch runs in $WORK, not in golden
    git -C "$WORK/$r" switch -c smp origin/xpack-development
done
```

The golden repositories are **only read**. Every write (`fetch` , `switch` , `commit`) runs in `$WORK` .

12.2 Building one commit: four git-native ways to lift content

Need	Command	Note
New file, whole	<code>git checkout golden/smp -- <path></code>	Then <code>git restore --staged <path></code> if other paths are staged. Commit with an explicit path list (step 3 below).
Part of a modified file	<code>git checkout -p golden/smp -- <path></code>	Interactive: answer <code>y</code> only for the hunks of this commit's subject.
Hunk needs editing (two subjects in one hunk)	<code>git checkout -p</code> then <code>e</code>	Or edit in the editor. The diff to golden for the remaining subject stays visible with <code>git diff golden/smp -- <path></code> .
Content from the devices repository	<code>git -C ../micro-os-plus-iii-devices show e8e39d5:<path> > <new path></code>	Or <code>git read-tree --prefix=soc/rp2350/ -u <tree-ish></code> after fetching devices as a remote (§14.2).

12.3 The per-commit cycle

1. BRANCH `git switch -c pr/<ID>-<slug>` (stacked on the previous pr/ branch)
2. LIFT one subject only (§12.2)
3. COMMIT `git add <exact paths> && git commit` (message template §12.4)
4. GATE xpm action(s) of the commit's stage (§13.3); a red gate is fixed by amending, never by a follow-up
5. TAG `git tag pr-<ID>` (optional, for bookkeeping)

12.4 Commit message template

```
<type>(<scope>): <imperative summary, at most 72 chars>

Problem:  what is wrong or missing in xpack-development
Failure:  the visible symptom (error text, wrong result, hang)
Change:   what this commit does, file by file if more than one
Scope:    single-core | SMP-guarded | test | build | docs
Gate:     xpm run <action> -> <N>/<N> passed
Group:    <group name> (only for R6 groups)
Source:    golden smp <short sha or subject> | devices@e8e39d5:<path>
Legacy:    PR #<n> of Part I (when it maps to one)

Co-Authored-By: ...
```

`type` is one of `fix`, `feat`, `test`, `build`, `docs`, `chore`, `refactor`.

12.5 From commits to pull requests

Each commit is on its own `pr/<ID>--<slug>` branch, stacked on the previous one in the same repository. A PR targets `xpack-development` for the first commit and the previous `pr/` branch for the others (as in §9.2). After a merge, the next PR is retargeted. Because of R1, R2 and R7, any Phase 1 commit can also be cherry-picked onto `xpack-development` on its own.

13. Global Ordering, Stages and Gates

13.1 Stages

Stage	Content	Repositories	Ends with gate
S0	Dissolve <code>devices</code> into the arch repos, retire <code>devices</code>	P, C, A32, A64, D	G0
S1	Single-core corrections and improvements (Part I PR #1–#22, plus new ones)	K (+C, P for the hrclock group)	G1 after every commit
S2	SMP core: kernel SMP, POSIX host SMP, Cortex-M SMP contract	K, P, C	G1 after every commit; G2 at the end
S3	New silicon and new architectures: M33, RP2350, AArch32, AArch64 ports	C, A32, A64	Each port commit builds its port library
S4	Test framework: kernel harness, platforms, port test builders, then one commit per test	K, P, C, A32, A64	G1 + G3 (growing)
S5	Documentation, one commit per document	K (+ port READMEs)	None (docs only)
S6	Final parity and the full test run	all	G-final

13.2 Cross-repository order (R6)

```
S0:  P01-P03 -> C01-C03 -> A32-01..03 -> A64-01..02 -> D01 (retire devices)
S1:  K01 ... K29   (group G-hrclock = K28 + C04 + P04; gate after the last of the three)
S2:  K30-K38 (kernel SMP, guarded) -> P05-P11 (host SMP) -> C05-C07 (Cortex-M SMP contract)
S3:  C08-C12 (M33, RP2350) | A32-04..12 | A64-04..11      (the three columns are independent)
S4:  K40-K47 (harness) -> P20 / C20 / A32-20 / A64-20 (port test builders and boards)
      -> K50.. (one platform per commit) -> one commit per test in each port
S5:  K90..
```

13.3 Gates (all of them are xpm actions in `micro-os-plus-iii/tests/package.json`)

Once per workspace:

```
cd "$WORK/micro-os-plus-iii-posix-arch" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-cortexm"    && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch32"    && xpm install && xpm link  # from S3
cd "$WORK/micro-os-plus-iii-aarch64"    && xpm install && xpm link  # from S3
cd "$WORK/micro-os-plus-iii/tests"      && xpm run install-all
```

Gate	Command (in <code>micro-os-plus-iii/tests</code>)	Pass criterion
G0	<code>xpm run test-native-cmake && xpm run test-cortex-cmake</code>	Same result as on <code>xpack-development</code> plus PR #1. Note that the base is red on <code>-Werror=extra-semi</code> in <code>dirent.h</code> until K01.
G1	<code>xpm run test-native-cmake</code> + <code>xpm run test-cortex-cmake</code> (the two may run in parallel)	24 configurations × 3 suites = 72/72 with the <code>xpack-development</code> harness. After the harness switch (K44–K46), all <code>ctest</code> entries of the 24 configurations pass, and the new count is written into the <code>Gate:</code> line.
G2	The G1 commands, then the SMP builds of stage S2 through <code>xpm run test-native-cmake</code> (posix-arch <code>test/native/smp-*</code> after P21+)	G1 green, and the SMP host tests pass
G3	<code>xpm run test-<platform>-cmake</code> for the platform the commit touches	All non- <code>hwd</code> tests of the platform pass (<code>ctest -LE hwd</code>)
G-final	<code>xpm run test-all</code>	native + cortex + <code>test-smp-cmake</code> all green
Unifdef review (S2 only, optional)	<code>git diff HEAD~1 \\\ unifdef -UOS_USE_SMP_SCHEDULER</code> (review aid)	No single-core change

`hwd` tests (real hardware: pico2, weact, nucleo, Raspberry Pi, Luckfox) are built by G3 but run only on a bench, through `xpm run test-<...>-hwd --config <cfg>` . Their result is recorded in the commit message when it is available.

14. Stage S0 — Dissolving `devices` with Commits Only

14.1 Source layout → destination layout

<code>devices@e8e39d5</code>	posix-arch	cortexm	aarch32	aarch64
<code>include/</code> (<code>sd.hpp</code> , <code>fatfs.hpp</code> , <code>fatfs_hw.hpp</code> , <code>usb_dwc2.hpp</code>)	<code>drivers/include/</code>	—	<code>drivers/include/</code>	<code>drivers/include/</code>
<code>src/</code> (<code>sd.cpp</code> , <code>fatfs.cpp</code> , <code>usb_dwc2.cpp</code>)	<code>drivers/src/</code>	—	<code>drivers/src/</code>	<code>drivers/src/</code>
<code>fatfs/</code> (FatFs R0.15 + <code>diskio_sd.cpp</code> , <code>fatfs_hw.cpp</code>)	<code>drivers/fatfs/</code>	—	<code>drivers/fatfs/</code>	<code>drivers/fatfs/</code>
<code>soc/native/</code>	<code>soc/native/</code>	—	—	—
<code>soc/stm32f4xx/</code>	—	<code>soc/stm32f4xx/</code>	—	—
<code>soc/rp2350/</code>	—	<code>soc/rp2350/</code>	—	—
<code>soc/bcm2837/</code>	—	—	<code>soc/bcm2837/</code>	<code>soc/bcm2837/</code>
<code>soc/rk3506/</code> (+ <code>sdcard.md</code>)	—	—	<code>soc/rk3506/</code>	—

14.2 Commit list

To import without scripts, use `git fetch ../micro-os-plus-iii-devices smp:devices-src` once per arch repo, then `git read-tree --prefix=<dest>/ -u devices-src:<src>` for each row. This only stages the added files; commit each one separately. The branch `devices-src` is local and is deleted after S0.

ID	Repo	Commit	Files	Gate
P01	posix-arch	<code>feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers</code>	<code>drivers/ {include,src,fatfs }/**</code>	builds (not yet referenced)
P02	posix-arch	<code>feat(soc): import the native host- file SD backend</code>	<code>soc/native/**</code>	idem
P03	posix-arch	<code>build(cmake): define the dissolved devices targets locally</code>	<code>CMakeLists.txt</code> : drop <code>UOS_DEVICES_DIR</code> and its check, add <code>micro-os- plus::devices- hostfile</code> and <code>micro-os- plus::devices</code> (§3.3 block)	G0
C01	cortexm	<code>feat(soc): import the STM32F4xx SoC support</code>	<code>soc/stm32f4xx/**</code>	builds
C02	cortexm	<code>feat(soc): import the RP2350 SoC support</code>	<code>soc/rp2350/**</code>	builds
C03	cortexm	<code>build(cmake): define the dissolved SoC targets locally</code>	<code>CMakeLists.txt</code> : drop <code>UOS_DEVICES_DIR</code> , add <code>soc- stm32f411xe</code> , <code>soc- stm32f412rx</code> , <code>soc- rp2350</code>	G0
A32-01	aarch32	<code>feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers</code>	<code>drivers/**</code>	— (the repo has no build yet)
A32-02	aarch32	<code>feat(soc): import the BCM2837 (Raspberry Pi 3 / Zero 2 W) SoC support</code>	<code>soc/bcm2837/**</code>	—
A32-03	aarch32	<code>feat(soc): import the RK3506 (Luckfox Lyra) SoC support</code>	<code>soc/rk3506/**</code>	—
A64-01	aarch64	<code>feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers</code>	<code>drivers/**</code>	—
A64-02	aarch64		<code>soc/bcm2837/**</code>	—

ID	Repo	Commit	Files	Gate
		<code>feat(soc): import the BCM2837 SoC support</code>		
D01	devices	<code>docs: retire the repository -- its contents moved to the architecture ports</code>	<code>README.md</code> only (a table of the moves in §14.1)	—

Each body has the trailer `Source: devices@e8e39d5:<src path>`. The devices history stays reachable in the archived repository, so a subtree merge is not needed. After D01: archive `dan-maio/micro-os-plus-iii-devices` on GitHub (Settings → Archive, an administrative action, not a commit), then delete the local clone (`rm -rf "$WORK/micro-os-plus-iii-devices"`). From here on, no CMake file may mention `micro-os-plus-iii-devices` (`git grep -n micro-os-plus-iii-devices` is empty in all five repositories).

15. Kernel `micro-os-plus-iii` — Commit List

15.1 Stage S1: single-core corrections and improvements (gate G1 after each)

ID	Commit subject	Files	Legacy
K01	<code>fix(posix): ISO C conformant DIR in dirent.h</code>	<code>include/cmsis-plus/posix/dirent.h</code>	PR #1
K02	<code>fix(utils): list iterators call node_>next()/prev()</code>	<code>include/cmsis-plus/utils/lists.h</code>	PR #2
K03	<code>fix(posix-io): match newlib's syscall return types in the weak aliases</code>	<code>include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h</code>	PR #3
K04	<code>fix(rtos): emit thread::suspend() out of line</code>	<code>src/rtos/os-thread.cpp</code> (that hunk only)	PR #4
K05	<code>fix(libc): declare timegm() only when the C library does not</code>	<code>src/libc/stdlib/timegm.c</code>	new (was hidden in step01)
K06	<code>fix(memory): block_pool asserts on a null result, not a valid one</code>	<code>src/memory/block-pool.cpp</code>	new (was hidden in step02)
K07	<code>fix(estd): polymorphic_allocator copy keeps its resource</code>	<code>include/cmsis-plus/estd/memory_resource</code>	new (was hidden in step02)
K08	<code>feat(memory): usable size for the RTOS allocators, overflow-checked calloc()</code>	<code>rtos/os-memory.h</code> , <code>src/rtos/os-memory.cpp</code> , <code>memory/first-fit-top.h</code> , <code>src/memory/first-fit-top.cpp</code> , <code>src/memory/lifo.cpp</code> , <code>src/libc/stdlib/malloc.cpp</code>	PR #5
K09	<code>feat(libcpp): C++17 aligned operator new/delete</code>	<code>src/libcpp/new.cpp</code>	PR #6
K10	<code>fix(libcpp): error categories with static storage</code>	<code>src/libcpp/system-error.cpp</code>	PR #7
K11	<code>fix(libcpp): steady_clock::now() without 64-bit overflow</code>	<code>src/libcpp/chrono.cpp</code>	PR #8
K12	<code>fix(rtos): CMSIS-RTOS osTimerCreate() defaults to one-shot</code>	<code>src/rtos/os-c-wrapper.cpp</code> (hunk)	PR #9 a
K13	<code>fix(rtos): CMSIS-RTOS mutex delete through the polymorphic type</code>	idem (hunk)	PR #9 b
K14		idem (hunk)	PR #9 c

ID	Commit subject	Files	Legacy
	<code>fix(rtos): widen millisec before scaling in the C wrapper</code>		
K15	<code>fix(posix-io): serialize the file descriptor table</code>	<code>src/posix-io/file-descriptors-manager.cpp</code>	PR #10
K16	<code>fix(posix-io): lock the deferred file-system and socket lists</code>	<code>posix-io/file-system.h</code> , <code>posix-io/net-stack.h</code>	PR #11
K17	<code>fix(posix-io): block_device sector-size check was inverted</code>	<code>src/posix-io/block-device.cpp</code>	PR #12
K18	<code>feat(arm): ARMv8-M semihosting trap</code>	<code>include/cmsis-plus/arm/semihosting.h</code>	PR #13
K19	<code>feat(startup): ARMv8-M exception handlers and fault dump</code>	<code>src/startup/exception-handlers.c</code> , <code>include/cmsis-plus/cortexm/exception-handlers.h</code>	PR #14
K20	<code>fix(semihosting): fstat() keeps an explicit mode; weak console mirror hook</code>	<code>src/semihosting/c-syscalls-semihosting.cpp</code>	PR #15 (split into two commits if the hunks are independent)
K21	<code>fix(rtos): run timer callbacks outside the critical section</code>	<code>src/rtos/internal/os-lists.cpp</code> , <code>src/rtos/os-timer.cpp</code> (hunks)	PR #16 a
K22	<code>fix(rtos): periodic timer drift catch-up guard</code>	<code>src/rtos/os-timer.cpp</code> (hunks)	PR #16 b
K23	<code>fix(rtos): mutex priority ceiling before ownership, highest boost among waiters</code>	<code>src/rtos/os-mutex.cpp</code>	PR #17
K24	<code>feat(rtos): thread state destroying, and an idle reaper that waits for it</code>	<code>rtos/os-c-decls.h</code> , <code>rtos/os-thread.h</code> , <code>src/rtos/os-idle.cpp</code> , <code>estd/condition_variable</code> (step09 hunks)	PR #18
K25	<code>fix(rtos): condition variable without lost signals</code>	<code>src/rtos/os-condvar.cpp</code> , <code>rtos/os-condvar.h</code> , <code>diag/instrumentation.h</code> (<code>SUSPEND_CAUSE_CONDVAR</code>), <code>estd/condition_variable</code> (step10 hunks)	PR #19
K26	<code>fix(estd): std::thread::join() waits on the native</code>	<code>src/libcpp/thread-cpp.h</code> , <code>estd/thread_internal.h</code>	PR #20

ID	Commit subject	Files	Legacy
	handle, keeps its functor		
K27	fix(rtos): message queue reschedules after a wake-up	src/rtos/os-mqueue.cpp	PR #21
K28	feat(rtos): high-resolution clock uses the port's hardware counter	rtos/os-decls.h (hunk), src/rtos/os-clocks.cpp	PR #22 K, Group G-hrclock with C04, P04
K29	test(rtos-apis): 512 KiB RTOS arena for the FatFs leg, and a switch to leave FatFs out	tests/sources/rtos-apis/include/cmsis-plus/os-app-config.h , tests/sources/rtos-apis/src/main.cpp (two commits, K29a/K29b, if you prefer one subject each)	new

Also in S1, one commit each, harness corrections that are independent of SMP: | ID | Commit subject | Files | ---|---|---| | K29c | fix(test): enable the FPU when the compiler targets one (__ARM_FP) | tests/device-qemu-cortexm/src/system-cortexm.c | | K29d | docs(test): the AN385 flash comment says 128M | tests/device-qemu-cortexm/linker-scripts/mem-mps2-an385.ld , ...an386.ld |

15.2 Remaining golden kernel changes and their commits (completeness check)

Every kernel file that differs between xpack-development and golden smp belongs to exactly one row of §15.1, §15.3, §15.4, §15.5 or §15.7, or to the intended differences in §19.2. When a file appears in two rows (for example os-thread.cpp in K04 and K35), each row takes only its own hunks.

15.3 Stage S2: kernel SMP core (guarded by `OS_USE_SMP_SCHEDULER`, gate G1 after each)

ID	Commit subject	Files (SMP hunks only)
K30	feat(port): smp-common -- the SMP declarations a thin port shares	port/smp-common/README.md , port/smp-common/cmsis-plus/rtos/ port/os-decls.h
K31	feat(rtos): SMP_NO_OWNER and the per-CPU thread context	rtos/os-c-decls.h , rtos/os-decls.h
K32	feat(rtos): one current thread per CPU	rtos/os-sched.h
K33	feat(rtos): thread CPU affinity and per-core run state	rtos/os-thread.h
K34	feat(rtos): SMP scheduler -- per-CPU ready lists, recursive kernel lock, no switch while holding it	src/rtos/os-core.cpp
K35	feat(rtos): thread migration and the five-stage stack-pointer claim/publish	src/rtos/os-thread.cpp
K36	feat(rtos): wake a thread's CPU with an IPI when it unblocks elsewhere	src/rtos/os-thread.cpp (IPI hunks; Part I PR #26 kernel part)
K37	feat(rtos): one idle thread per CPU	src/rtos/os-idle.cpp
K38	feat(rtos): pin the main thread to CPU 0	src/rtos/os-main.cpp
K39	test(validator): use glibc's ucontext, not libucontext	tests/sources/cmsis-os-validator/include/cmsis-plus/os-app-config.h (lands right after P05, see R6)

15.4 Stage S4a: build and harness infrastructure (gate G1 after each)

ID	Commit subject	Files
K40	<code>build(cmake): cross toolchain profiles for arm-none-eabi, aarch64-none-elf, native</code>	<code>cmake/toolchains/*.cmake</code>
K41	<code>build(cmake): uos-app.cmake -- one call builds an application against a port</code>	<code>cmake/uos-app.cmake</code>
K42	<code>build(cmake): modular kernel targets iii-core, iii-posix-io, port-smp-decls, test-support</code>	<code>CMakeLists.txt</code> (keep <code>micro-os-plus::iii</code> as the umbrella alias)
K42b	<code>build(cmake): fail the configure on a source compiled twice (optional)</code>	<code>tools/verify-no-duplicate-sources.py</code> plus its hook in <code>CMakeLists.txt</code> . If omitted, K42 has no hook.
K43a	<code>test: the verdict contract shared by every board (hw_result, board contract)</code>	<code>test_smpl/include/hw_result.hpp</code> , <code>test_smpl/src/board-contract.cpp</code>
K43b	<code>test: host runner for the port test applications</code>	<code>test_smpl/run-host.sh</code>
K43c	<code>test: QEMU runner for the port test applications</code>	<code>test_smpl/run-qemu.sh</code>
K43d	<code>test: hardware runner and the OpenOCD configs that need no power cycle</code>	<code>test_smpl/run-hw.sh</code> , <code>test_smpl/no-power-cycle/*</code>
K44	<code>test(cmake): find the ports as siblings (UOS_<PORT>_DIR), with the xpack link as fallback</code>	<code>tests/cmake/tests-main.cmake</code> , <code>tests/CMakeLists.txt</code>
K45	<code>test(native): run posix-arch's own test applications</code>	<code>tests/platforms/native/**</code> (after P20, R6)
K46a–d	<code>test(qemu-cortex-m0 m3 m4f m7f): build against the cortexm port targets</code>	<code>tests/platforms/qemu-cortex-m{0,3,4f,7f}/**</code> , one commit each (after C20)
K47	<code>test(qemu): MPS2 AN505/AN521 memory maps and ARMv8-M vectors</code>	<code>tests/device-qemu-cortexm/linker-scripts/mem-mps2-an505.ld</code> , <code>mem-mps2-an521.ld</code> , <code>src/vectors-cortexm.c</code> , <code>include/cmsis_device.h</code> , <code>src/exception-handlers.cpp</code>
K48	<code>test(nucleo-f411re): build against the dissolved soc-stm32f411xe target</code>	<code>tests/platforms/nucleo-f411re/**</code>

15.5 Stage S4b: new test suite (one commit per test)

ID	Commit subject	Files
K49	test: fp-switch -- floating-point context integrity under 1 ms preemption	tests/sources/fp-switch/**

15.6 Stage S4c: platforms (one commit per platform; gate G3 for that platform)

Each commit adds the platform directory **and** in `tests/package.json` its `*-cmake-gcc-{debug,release}` configurations, its `test-<platform>-cmake` action, and **one line** in `test-smp-cmake`. The first of these commits also creates `test-smp-cmake` and appends it to `test-all`. Configuration templates that several platforms share (`aarch32-actions`, `aarch32-dependencies`, `aarch64-*`, `arm-cmsis-core-dependencies`, `clang-gcc14-properties`) land with the first platform that uses them.

ID	Platform	Needs (R6)	QEMU in <code>test-smp-cmake</code>
K50	<code>2xcortex-m33</code> (MPS2 AN521, 2 × M33)	C08, K47	yes
K51	<code>pico2-1cpu</code> (RP2350 code at NCPU=1 on QEMU)	C09	yes
K52	<code>cortexm-pico2</code>	C21 + pico2 tests	yes
K53	<code>cortexm-pico2-rp2350b-psram</code>	C22	yes
K54	<code>cortexm-pico2-pizero</code>	C23	no (build and <code>hwd</code> only)
K55	<code>cortexm-nucleof411</code>	C24	no
K56	<code>cortexm-weactf411</code>	C25	no
K57	<code>cortexm-weactf412</code>	C26	no
K58	<code>aarch32-rpi-zero-2w</code> (raspi3b, 4 × A53)	A32-20, A32-21	yes
K59	<code>aarch32-rpi3b</code>	A32-22	yes
K60	<code>aarch64-rpi-zero-2w</code>	A64-20, A64-21	yes
K61	<code>aarch64-rpi3b</code>	A64-22	yes
K62	<code>aarch32-luckfox-lyra</code> (RK3506, 3 × A7; hardware only)	A32-23	no (<code>build-aarch32-luckfox-lyra-cmake</code>)
K63	<code>test(package): Debug/Release aliases and the clang 13-15 native configurations</code>	—	—

15.7 Stage S5: documentation (one commit per document)

One commit for each `docs/*.md` (18 files), for each `docs/smp-integration/*.md` together with its own `diagrams/*` (24 documents), for each `docs/tests/*` group, for `docs/specs/*`, for `README-DEVELOPER.md`, `tests/README.md` and `tests/TO-CHECK.md`, and for `port/smp-common/README.md` if it is not in K30. `.clang-format` and the `.gitignore` additions are one `chore` commit each.

16. `micro-os-plus-iii-posix-arch` — Commit List

16.1 S0 and S1

P01–P03 (§14.2). **P04** `feat(port): high-resolution counter from CLOCK_MONOTONIC` belongs to group G-hrclock with K28.

16.2 S2: the host as a multi-core machine (one feature per commit; later golden fixes folded in by R5)

ID	Commit subject	Files	Golden fixes folded in
P05	<code>fix(port): header hygiene for GCC < 14 and clang -Weverything</code>	<code>include/cmsis-plus/rtos/port/os-{c-decls,decls,inlines}.h</code> (non-SMP hunks)	"build with GCC < 14 (__has_feature)", Fixes A/B/I
P06	<code>feat(port): SMP port contract -- per-thread CPU, per-CPU ISR state, kernel lock</code>	the same headers (SMP hunks)	"block the tick before reading the CPU id in locked()", "never read the CPU id through a cached thread pointer"
P07	<code>feat(port): host CPUs are pthreads, with per-CPU timers and signal IPIs</code>	<code>include/host_cpu.hpp</code> , <code>src/host_cpu.cpp</code> , <code>CMakeLists.txt</code> line	"drop the dead timer lock; host CPU tick split", Fixes C/E/G
P08	<code>feat(port): host exception dispatch and the per-core boot stage</code>	<code>include/exception_handler.hpp</code> , <code>src/exception_handler.cpp</code> , <code>CMakeLists.txt</code> line	"handler mode for the validator's SIGUSR1 shim", Fix F
P09	<code>feat(port): the host free store</code>	<code>src/free-store.cpp</code> , <code>CMakeLists.txt</code> line	—
P10	<code>feat(port): SMP start-up and signal-driven context switch</code>	<code>src/rtos/os-core.cpp</code>	"errno is the μ OS++ thread's, not the host thread's"
P11	<code>feat(port): the board verdict contract on the host</code>	<code>include/hw_result.hpp</code> , <code>src/board-contract.cpp</code> , <code>CMakeLists.txt</code> line	—
P12	<code>feat(port): sanitizer support for the native board</code>	<code>CMakeLists.txt</code> (hunk)	—
P13	<code>build(cmake): resolve the kernel as a sibling under either directory name</code>	<code>CMakeLists.txt</code> (hunk)	"support micro-os-plus-iii directory name"
P14	<code>tools: a probe showing why TSan fiber annotations cannot work here</code> (optional)	<code>tools/tsan-fiber-probe.c</code>	—

16.3 S4: tests (P20 first, then one commit per test; gate `xpm run test-native-cmake`)

ID	Commit	Files
P20	<code>test: the port test builder and the native board</code>	<code>test/CMakeLists.txt</code> , <code>test/run.sh</code> , <code>test/boards/native/**</code> , <code>test/native/include/**</code> , <code>test/native/src/**</code>
P21	<code>test(native): rtos-apis at OS_NCPU=1</code>	<code>test/native/rtos-apis/**</code>
P22	<code>test(native): mutex-stress</code>	<code>test/native/mutex-stress/**</code>
P23	<code>test(native): flatfs-test -- append to a file created empty</code>	<code>test/native/flatfs-test/**</code>
P24	<code>test(native): mutex-ceiling-test -- a refused protect lock leaves nothing</code>	<code>test/native/mutex-ceiling-test/**</code>
P25	<code>test(native): smp_test0 ... P29 smp_test4</code>	one directory per commit
P30	<code>test(native): smp-num-test</code>	
P31	<code>test(native): smp-mat-test (N=500, B=50)</code>	
P32	<code>test(native): smp-pipeline-test</code>	
P33	<code>test(native): smp-pro-cons-test</code>	
P34	<code>test(native): smp-mutex-stress -- SMP leg with per-core checks</code>	
P35	<code>test(native): smp-rtos-apis -- SMP leg with per-core checks</code>	

All of them use `std::atomic` for counters shared between cores (golden fix "convert multi-core volatile counters to std::atomic", folded in by R5).

17. `micro-os-plus-iii-cortexm` — Commit List

17.1 S0 and S1

C01–C03 (§14.2). **C04** `feat(port): high-resolution counter, with the SysTick pending-overflow check` belongs to group G-hrclk.

17.2 S2: Cortex-M SMP contract (single-core Cortex-M builds unchanged)

ID	Commit	Files	Golden fixes folded in
C05	<code>feat(port): Cortex-M SMP contract -- per-core lock state, recursive kernel lock</code>	<code>include/cmsis-plus/rtos/port/os-{c-decls,decls,inlines}.h</code>	"mask IRQs before reading the core id in scheduler::locked()", "publish scheduler unlock before releasing the lock"
C06	<code>feat(port): PendSV switch with stack-pointer claim/publish</code>	<code>src/rtos/os-core.cpp</code>	"unpublish the stack pointer on an identity reschedule", "release klock on null thread in switch_stacks", "save the FPU context on a switch"
C07	<code>fix(cortexm): scheduler guards, semihosting clobbers, linker sections</code>	the hunks of that golden commit that touch upstream code	—

17.3 S3: new silicon

ID	Commit	Files
C08	<code>feat(port): generic Cortex-M33 (SSE-200) dual-core port</code>	<code>include-m33/**</code> , <code>src/rtos/os-core-m33.cpp</code> , <code>CMakeLists.txt</code> target
C09	<code>feat(port): RP2350 dual-core port -- SIO spinlock, FIFO IPI, core 1 launch</code>	<code>include-rp2350/**</code> , <code>src/rtos/os-core-rp2350.cpp</code> , <code>CMakeLists.txt</code> target (with "release the 32 SIO spinlocks at reset" folded in)
C10	<code>feat(libc): getentropy() for the bare-metal ports</code>	<code>src/libc/getentropy.c</code> , <code>CMakeLists.txt</code> line
C11	<code>feat: a semihosting exit that reports the test verdict</code>	<code>src/semihosting-exit.cpp</code> , <code>CMakeLists.txt</code> line
C12	<code>build(cmake): resolve the kernel as a sibling under either directory name</code>	<code>CMakeLists.txt</code> (hunk)

17.4 S4: tests (C20 first, then one board per commit, then one test per commit)

ID	Commit	Files
C20	test: the port test builder, QEMU and hardware dispatch	test/CMakeLists.txt, test/qemu.sh, test/hw.sh, test/boards/shared/**
C21	test(pico2): the Pico 2 board	test/boards/pico2/**, test/pico2/{CMakeLists, tests.cmake, include, src}
C22	test(pico2-rp2350b-psram): the RP2350B PSRAM board	test/boards/pico2-rp2350b-psram/**, test/pico2-rp2350b-psram/tests.cmake
C23	test(pico2-pizero): the Pico 2 on Pi Zero carrier	test/boards/pico2-pizero/**
C24	test(nucleof411): the NUCLEO-F411RE board under the SMP scheduler at one CPU	test/boards/nucleof411/**
C25	test(weactf411): the WeAct F411 board	test/boards/weactf411/**
C26	test(weactf412): the WeAct F412 board	test/boards/weactf412/**

Then one commit per test directory (`test(<board>): <test>`): - **pico2**: cmsis-os-validator, cmsis-os-validator-ram, exc-test, fp-switch, mutex-stress, mutex-stress-ram, rtos-apis, rtos-apis-ram, sc-test-ko, smp-mat-test, smp-mat-test-ram, smp-test0 ... smp-test5, smp-test-ko, smp-test-usb-cdc-acm, smp-test-usb-hid (25 commits). - **pico2-rp2350b-psram**: exc-test, fp-switch, sc-test-ko, smp-mat-test, smp-test0 ... smp-test5, smp-test-ko, smp-test-nested, smp-test-nested-clock, smp-test-nested-clock_200, smp-test-nested-clock_250 (16). - **pico2-pizero**: exc-test, psram-exec, psram-mat-test-250, sc-test-ko, smp-mat-test, smp-test0 ... smp-test5, smp-test-ko, smp-test-usb-cdc-acm, smp-test-usb-hid (14). - **nucleof411**, **weactf411**, **weactf412**: cmsis-os-validator, mos-test1, mutex-stress, rtos-apis; plus weactf411 spi-pipeline and weactf412 uart-test1 (14).

18. `micro-os-plus-iii-aarch32` and `micro-os-plus-iii-aarch64` — Commit Lists

On `xpack-development` both repositories contain only `.gitignore`, `.npmignore`, `LICENSE`, `README.md` and `package.json`. Everything is therefore an **add**, plus one `.gitignore` update.

18.1 S3: the port (each commit builds `micro-os-plus::aarch32` / `::aarch64` from the files present so far, R8)

AArch32	AArch64	Commit	Files
A32-04	A64-03	<code>chore: ignore every build* directory</code>	<code>.gitignore</code>
A32-05	A64-04	<code>build(cmake): the port project, the dissolved devices target, sibling kernel</code>	<code>CMakeLists.txt</code> (initially only the targets whose sources exist, plus <code>micro-os-plus::devices</code> from §3.3)
A32-06	A64-05	<code>feat(port): device header, application config and port declarations</code>	<code>include/cmsis_device.h</code> , <code>include/cmsis-plus/os-app-config.h</code> , <code>include/cmsis-plus/rtos/port/os-{c-decls,inlines}.h</code>
A32-07	A64-06	<code>feat(port): MMU set-up (A64: with <code>TCR_EL1.EPD1</code> folded in)</code>	<code>include/mmu.hpp</code>
A32-08	A64-07	<code>feat(port): exception vectors and handlers</code>	<code>include/exception_handler.hpp</code> , <code>src/exception_handler.cpp</code> , <code>src/handlers.cpp</code>
A32-09	A64-08	<code>feat(port): ARM generic timer and the high-resolution counter (<code>CNTPCT</code> / <code>CNTPCT_EL0</code>)</code>	<code>include/timer_arm.hpp</code>
A32-10	A64-09	<code>feat(port): context switch with stack-pointer claim/publish</code>	<code>include/port_ctx.hpp</code> , <code>src/context_switch.cpp</code> , <code>src/rtos/os-core.cpp</code> (MPIDR Aff0 mask folded in)
A32-11	A64-10	<code>feat(port): secondary-core boot</code>	<code>src/smp_secondary.cpp</code>
A32-12	A64-11	<code>feat(port): semihosting console and exit</code>	<code>include/semihosting.hpp</code> , <code>src/semihosting-exit.cpp</code> (A32 only)

18.2 S4: tests

AArch32	AArch64	Commit	Files
A32-20	A64-20	test: the port test builder, QEMU shim and hardware dispatch	test/CMakeLists.txt , test/qemu.sh , test/hw.sh
A32-21	A64-21	test(rpi-zero-2w): the Zero 2 W board	test/boards/rpi-zero-2w/** , test/rpi-zero-2w/{tests.cmake,include,src}
A32-22	A64-22	test(rpi3b): the Pi 3 B board (shares the Zero 2 W sources)	test/boards/rpi3b/** , test/rpi3b/{tests.cmake,include,src}
A32-23	—	test(luckfox-lyra): the Luckfox Lyra board (RK3506, hardware only)	test/boards/luckfox-lyra/** , test/luckfox-lyra/{include,src}

Then one commit per test directory, for each of `rpi-zero-2w` and `rpi3b` in both repositories: `cmsis-os-validator`, `mutex-stress`, `rtos-apis`, `sd_test`, `smp-mat-sdcard-test`, `smp-mat-test`, `smp-num-test`, `smp-pipeline-test`, `smp-pro-cons-test`, `smp_test0` ... `smp_test4`, `usb_test` (18 per board, 72 commits in all).

For `luckfox-lyra` (AArch32 only): `sd_test`, `smp-mat-sdcard-test`, `smp-mat-test`, `smp-num-test`, `smp-pipeline-test`, `smp-pro-cons-test`, `smp_test0` ... `smp_test7`, `smp_test_int`, `smp_test_int2` ... `smp_test_int5` (19 commits).

The kernel platform for a board (K58–K62) lands after the board commit and at least one test. Each further test commit makes one more `ctest` entry appear in `xpm run test-<platform>-cmake`.

19. Final Parity and the Full Test Run

19.1 Parity check (read-only against golden)

```
for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-iii-posix-arch \
    micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
    git -C "$WORK/$r" diff --stat golden/smp smp
done
```

19.2 Intended differences (everything else is a defect to fix with one more commit)

1. Files golden deletes but the rebuild keeps (R2): kernel `.github/`, `doxygen/`, `inspiration/`, `templates/`, `.vscode/`, `.settings/`, `CHANGELOG.md`, `CLAUDE.md`, `README.md`, `README-MAINTAINER.md`, `LICENSE` (not renamed), `package-lock.json`, and the `link-deps-native-cmake-`

`sys` action. Ports: `.npmignore`, `LICENSE`, `README*.md`, `CHANGELOG.md`, `NOTES.md`, `scripts/xpacks-helper.sh`, `.vscode/`, `.clang-format`, `posix-arch` `test/trace-*`.

2. Procedure tooling golden has but the rebuild does not (R3): `scripts/smp/**`, `tests/xpacks/**`, `README.pdf`, and the optional `tools/*` that were not taken.
3. Content the rebuild has but golden lacks: `soc/` and `drivers/` in `posix-arch`, `cortexm`, `aarch32` and `aarch64`. Golden's CMake references them, but they are absent from its tree.
4. Any version bump made as an optional `chore(release)` commit.

19.3 Full test run (µOS++ test framework only)

```
cd "$WORK/micro-os-plus-iii/tests"
xpm run install-all
xpm run test-all          # native (gcc11-14, clang16-19) + qemu-cortex-m0/m3/m4f/m7f + test-smp-cmake
```

`test-smp-cmake` runs `aarch32-rpi-zero-2w`, `aarch32-rpi3b`, `aarch64-rpi-zero-2w`, `aarch64-rpi3b`, `2xcortex-m33`, `pico2-1cpu`, `cortexm-pico2` and `cortexm-pico2-rp2350b-psram` on QEMU. Hardware-only platforms (`cortexm-pico2-pizero`, `cortexm-nucleof411`, `cortexm-weact*`, `aarch32-luckfox-lyra`, `nucleo-*`, `raspberrypi-pico`) are built by `test-all` where they are registered, and are run on the bench with their `hwd` actions.

19.4 Size of the series (planning estimate; the executed counts are in §20.1 and §22.1)

Repository	Commits (approx.)	Of which tests
micro-os-plus-iii	29 + 10 + 13 + 1 + 14 + ~45 docs ≈ 110	1 suite + 13 platforms
posix-arch	3 + 1 + 10 + 16 ≈ 30	16
cortexm	3 + 1 + 3 + 5 + 7 + 69 ≈ 88	69
aarch32	3 + 9 + 4 + 36 + 19 ≈ 71	55
aarch64	2 + 9 + 3 + 36 ≈ 50	36
devices	1 (retirement)	—

Every row is one pull request. Tests and documents are small and add-only, so they review fast and can be merged in bulk once their builder or platform commit has landed.

Part III — Execution Record, Step Summary and Complete Commit List

Part II was executed once from scratch on 2026-10-06. Fresh clones of `github.com/dan-maio/*` went into `/tmp`, with `~/Work2/micro-os-plus/*` as the read-only `golden` remote. This part records what

was done, how it differs from the Part II plan, and every commit that resulted. It is the reference for opening the pull requests.

20. Summary of All Steps: `xpack-development` (single CPU) → `smp` (SMP)

20.1 Starting point

Repository	Base (<code>origin/xpack-development</code>)	Result branch	Commits
<code>micro-os-plus-iii</code> (kernel, K)	<code>7f1ce5ca</code>	<code>smp</code>	125 (+ 34 documentation updates, §22.4)
<code>micro-os-plus-iii-posix-arch</code> (P)	<code>86a6a1f</code>	<code>smp</code>	24
<code>micro-os-plus-iii-cortexm</code> (C)	<code>687e975</code>	<code>smp</code>	86
<code>micro-os-plus-iii-aarch32</code> (A32)	<code>307c62f</code>	<code>smp</code>	64
<code>micro-os-plus-iii-aarch64</code> (A64)	<code>fe904c0</code> (GitHub; golden's base <code>366c957</code> is one commit older)	<code>smp</code>	43
<code>micro-os-plus-iii-devices</code> (D)	<code>smp</code>	retired (D01)	1
Total			343

Each commit carries the trailer `Commit-ID: <ID>` (`xpack-dev-smp.md` Part II) . Each commit also has a local branch `pr/<ID>` pointing at it, so commit n becomes the PR `pr/<ID>` → `pr/<previous ID>` (§12.5). Nothing has been pushed.

20.2 The steps, in the order they were committed

Step	Stage	Repositories and commit IDs	What the step does	Build check
1	Workspace	all	Clone from <code>dan-maio</code> , add the read-only <code>golden</code> remote, create <code>smp</code> from <code>origin/xpack-development</code> (§12.1). Install with <code>xpm install</code> / <code>xpm link</code> in each port, then <code>xpm run install-all</code> in <code>micro-os-plus-iii/tests</code> (§13.3).	<code>install-all</code> rc 0
2	S0 devices	P01–P03, C01–C03, A32-01..03, A64-01..02, D01	Dissolve <code>devices</code> . Its drivers (SD, flatfs, USB DWC2, FatFs) go to <code>drivers/</code> and its SoC support to <code>soc/<chip>/</code> in each port that uses them. The <code>micro-os-plus::devices*</code> / <code>soc-*</code> CMake targets are defined in the port. The devices repo gets one README commit that records where everything went.	native + cortex build
3	S1 single-core	K01–K29d (41 commits)	Upstream corrections and improvements, one per commit. They cover <code>dirent</code> , <code>lists</code> , the syscall aliases, <code>suspend</code> , the allocators, aligned <code>new</code> , <code>system_error</code> , <code>chrono</code> , the C API, POSIX I/O locking, block device, ARMv8-M semihosting and fault handlers, timers, the mutex ceiling and inheritance, the thread life cycle (<code>destroying</code> , <code>detach</code> , <code>join</code>), the condition variable, <code>std::thread</code> , mqueue, and the rtos-apis test arena.	build after each
4	S1 hrclock group			

Step	Stage	Repositories and commit IDs	What the step does	Build check
		K28a, K28 + C04, C04b + P04	The high-resolution clock reads the port's hardware counter. The kernel declares it, cortexm reports that it has no hardware counter and fixes <code>cycles_since_tick()</code> , and posix-arch reads <code>CLOCK_MONOTONIC</code> .	builds at the last commit of the group
5	S2 SMP core (kernel)	K30–K39b	Everything is guarded by <code>OS_USE_SMP_SCHEDULER</code> : <code>port/smp-common</code> , one current thread per CPU, CPU affinity, the SMP picker with the stack-pointer claim, the IPI wake-up, <code>join</code> / <code>kill</code> / reaper aware of other CPUs, main thread and CMSIS-RTOS v1 threads on CPU 0.	single-core builds unchanged
6	S2 host SMP	P05, P06	posix-arch becomes a multi-core host (one host thread per CPU, signal IPIs, per-CPU timers, exception dispatch) and gets the board verdict contract.	native build
7	S2 Cortex-M SMP	C05–C08	The SMP port contract on the generic Cortex-M core (SMP scheduler at one CPU), a <code>_getentropy()</code> stub, and a semihosting <code>_Exit()</code> that reports the verdict.	cortex build
8	S3 new silicon	C09, C10	A generic dual-core Cortex-M33 (SSE-200) port for QEMU, and the RP2350 dual-core port.	build
9	S3 new architectures	A32-04..11, A64-03..10	The AArch32 and AArch64 ports: declarations, MMU, vectors, generic timer, context switch, secondary-core entry, semihosting. Add-only;	— (no board yet)

Step	Stage	Repositories and commit IDs	What the step does	Build check
			nothing builds them until step 12.	
10	S4a harness	K40, K41, K43a–d, then group G-harness: K42, K44, K45, K46–m0..m7f + P20 + C20a/b, then K47	Toolchain profiles, <code>uos_add_app()</code> , the <code>test_smp1</code> verdict contract and runners, and modular kernel targets. Each port has a test builder that selects a board, and the kernel harness finds the ports as siblings. The native and QEMU Cortex-M platforms build against the port targets.	builds at the last commit of the group
11	S4b/c Cortex-M platforms	K49, K50–K57; C21–C27 boards; one commit per test	The fp-switch suite; platforms <code>2xcortex-m33</code> , <code>pico2-1cpu</code> , <code>cortexm-pico2</code> , <code>-rp2350b-psram</code> , <code>-pizero</code> , <code>-nucleof411</code> , <code>-weactf411</code> , <code>-weactf412</code> ; the boards pico2 (with TinyUSB in C21b–d), pico2-rp2350b-psram, pico2-pizero, nucleof411, weactf411 and weactf412; 63 test commits.	build per commit
12	S4 AArch32/AArch64 boards	A32-20, A32-21, A32-R3, A32-LL; A64-20, A64-21, A64-R3; K58–K62	Port CMake, board models (rpi-zero-2w, rpi3b, luckfox-lyra), and the kernel platforms <code>aarch32-rpi-zero-2w</code> , <code>aarch32-rpi3b</code> , <code>aarch32-luckfox-lyra</code> , <code>aarch64-rpi-zero-2w</code> , <code>aarch64-rpi3b</code> .	build per commit
13	S4 port tests	P21–P35, A32-T-, A32-R3-, A32-LL-, A64-T-, A64-R3-*	One commit per test directory, each adding one <code>ctest</code> entry: 15 native, 30 AArch32 Raspberry Pi, 20 Luckfox Lyra, 30 AArch64.	build per commit
14	S4 remaining harness			see §21.3

Step	Stage	Repositories and commit IDs	What the step does	Build check
		K39c, K48-nucleo-f411re/-f767zi/-h743zi/-raspberrypi-pico	The validator host build uses glibc's <code>ucontext</code> . The legacy hardware platforms label their tests <code>hwd</code> and link the modular targets.	
15	S5 docs and chores	K901–K935, K980–K984, P14, P36, C28, C29; after the run K985–K989 (§22.4)	One commit per document; <code>.clang-format</code> ; <code>.gitignore</code> additions; the TSan probe; the cortexm board README; this document.	—
16	S6 parity	all (read-only diff with <code>golden/smp</code>)	§21.4.	—
17	S6 full test	<code>micro-os-plus-iii/tests</code>	§21.5.	all green

21. Execution Record

21.1 Where the executed IDs differ from the Part II plan

Plan (Part II)	Executed	Reason
K05 <code>timegm()</code> declaration	dropped	Golden's version fails on glibc 2.44 with <code>-Werror=missing-prototypes</code> . The baseline <code>timegm.c</code> is kept (intended difference).
K20, K23, K24, K25, K26	split into K20a/b, K23a–c, K24a–e, K25a/b, K26a/b	R1: their golden hunks turned out to hold several independent subjects.
K21a (new)	<code>fix(rtos): errno in handler mode does not assert</code>	A separate golden subject in the K21 files.
K22 periodic-timer drift guard	folded into K21	It shares one hunk with the relink (stated in K21's <code>Legacy:</code> line).
K28a (new)	<code>os-decls.h</code> includes the C declarations again	The provider of K28 (R6).
K31–K39	renumbered by subject (see §22.2)	The golden SMP hunks group by behaviour (current thread, affinity, picker, IPI, join, kill, reaper, CPU 0) rather than by file.
K39b (new)	<code>port_cpu_id()</code> declarations tolerate the port's own	Needed by P05.
K42b duplicate-source check	not taken	It is a Python tool (R3).
K46a–d	K46-m0, -m3, -m4f, -m7f	Named by platform.
K48	four commits K48-nucleo-f411re/-f767zi/-h743zi/-raspberrypi-pico	One per legacy platform (R1).
K63 Debug/Release aliases, clang 13–15	not needed	<code>tests/package.json</code> is already byte-identical to golden after K62.
K90.. docs	K901–K935 (one per document) + K980–K984	§15.7.
P05–P13 host SMP	P05 (one commit) + P06	Golden's host-SMP files only compile together. The fixes (A–J, GCC < 14, errno) are folded in (R5).
C05–C12	C04b, C05–C10	<code>cycles_since_tick()</code> became its own fix. <code>getentropy/_Exit</code> came before the new silicon. The CMake sibling lookup was folded into C20b.
C20	C20a (nucleof411 board, needed by the builder) + C20b (builder)	R6.
C21–C26 boards	C21 (+C21b–d TinyUSB), C23–C27	pico2's USB stack is three commits. The pico2 test series is C22-*.
A32-05..12 / A64-04..11	A32-05..11 / A64-04..10, then A32-21 / A64-21 for CMake	Port CMake lands with the first board (R8: no target without sources).
A32-22/23, A64-22	A32-R3, A32-LL, A64-R3	Named by board.

21.2 Groups (built only at their last commit)

Group	Commits	Why
G-hrclock	K28a, K28, C04, C04b, P04	The kernel calls <code>clock_highres::has_hardware_counter()</code> , which each port defines.
G-harness	K42, K44, K45, K46-m0..m7f, P20, C20a, C20b	The kernel test harness switches from its own device sources to the port test builders, and neither side builds alone. The messages of the intermediate commits say so.

21.3 Builds

Every commit was built with `xpm run build --config <cfg>` in `micro-os-plus-iii/tests`, with uncommitted changes stashed, and was green, except: 1. The intermediate commits of the two groups (§21.2). 2. A32-01..11 and A64-01..10. They only add files, and nothing builds them until the board commits A32-21 / A64-21. 3. `nucleo-f411re`, `nucleo-f767zi`, `nucleo-h743zi` and `raspberrypi-pico` fail at configure: the golden cortexm CMake does not find the kernel through the xpacks path. The files are identical to golden, so this is a golden defect, reported here and not fixed in this series.

Build fixes beyond golden, each folded into the commit that introduced the code (R5): - warning suppressions in `system-error.cpp`, `first-fit-top.cpp`, `file-descriptors-manager.cpp`; - Fix H pragmas in `os-core.cpp` and `os-thread.cpp`; - the kernel `test-support` target falls back to `test_smpl/src/board-contract.cpp` (K42); - posix-arch fixes A–J (P05); - the AArch32 `micro-os-plus::soc-bcm2837` and `micro-os-plus::devices-rk3506` targets (A32-21); - the AArch64 `soc-bcm2837` target (A64-21).

21.4 Final parity with `~/Work2/micro-os-plus/*` (`golden/smp`, devices excluded)

- **Identical:** all cortexm sources and CMake, and `tests/package.json`.
- **Kept although golden deletes them (R2):** `.github/`, `doxygen/`, `inspiration/`, `templates/`, `.vscode/`, `.settings/`, `CHANGELOG.md`, `CLAUDE.md`, `README*.md`, `LICENSE` (not renamed), `package-lock.json`, the port `LICENSE/` `.npmignore/` `README.md`, posix-arch `test/trace-*`, and `scripts/xpacks-helper.sh`.
- **Golden content not taken (R3):** `scripts/smp/**`, `tools/verify-*`, `docs/**/*sh` and `*.py`, the diagram generators, `tests/xpacks/**`.
- **In the rebuild but not in golden:** `soc/` and `drivers/` in all four ports (the dissolved devices).
- **Build fixes:** listed in §21.3; plus `timegm.c` at baseline.
- **Other:** the aarch64 `package.json` description is the original one, and this document's Part II and Part III are new.

No other difference remains.

21.5 Full test run (μOS++ test framework, latest compilers, `sys` for native)

Action (in <code>micro-os-plus-iii/tests</code>)	Result
<code>xpm run install-all</code>	rc 0
<code>xpm run test-native-cmake-sys</code> (debug, release)	16/16 passed in each
<code>xpm run test-cortex-cmake</code> (qemu-cortex-m0/m3/m4f/m7f × gcc/clang × debug/release = 8 configs)	3/3 passed in each. The CMSIS validator reports 60/60.
<code>xpm run test-smp-cmake</code> : <code>aarch32-rpi-zero-2w</code> , <code>aarch32-rpi3b</code> (debug, release)	15/15 passed in each
<code>aarch64-rpi-zero-2w</code> , <code>aarch64-rpi3b</code> (debug, release)	15/15 passed in each
<code>2xcortex-m33</code> (debug, release)	4/4 passed in each
<code>pico2-1cpu</code> (debug, release)	4/4 passed in each
<code>cortexm-pico2</code> (debug, release)	6/6 passed in each
<code>cortexm-pico2-rp2350b-psram</code> (debug, release)	3/3 passed in each

All of them run on QEMU or on the host; `ctest -LE hwd` excludes the real-hardware tests. The hardware-only platforms are built but not run.

21.6 Verification replay from golden (2026-10-06)

To check that the procedure of Part II, as recorded here, rebuilds the `smp` branches, the whole series was executed again from scratch in an empty `/tmp`:

1. **Clone.** Each repository was cloned from the golden working copies in `~/Work2/micro-os-plus/`, branch `xpack-development`, with the documented bases: kernel `7f1ce5ca`, cortexm `687e975`, posix-arch `86a6a1f`, aarch32 `307c62f`, aarch64 `fe904c0`, devices `e8e39d5`. The golden repositories are the read-only `golden` remote; the kernel's `golden/smp` is the commit before this document's own golden commit.
2. **Install.** `xpm run install-all` in `micro-os-plus-iii/tests`, then `xpm install` and `xpm link` in each port (§13.3).
3. **Commits.** Every commit of §22 was made again in the same order, with the same lifts and hunk selections, and built after each commit as in the first run (`xpm run build --config <cfg>`).
4. **Result.** The same 375 commits (kernel 158, posix-arch 24, cortexm 86, aarch32 64, aarch64 43) plus devices D01, with the same Commit-IDs, order and subjects, and the same build result at every commit (the exceptions of §21.3 included). Only the SHAs differ, because the commit dates differ; the SHAs in §22.3 are those of this replay.
5. **Parity with golden `smp`.** Exactly the intended differences of §21.4, plus the documentation updates of §22.4. This document itself differs from the golden copy only in the SHA columns.
6. **Full test run** (latest compilers, `ctest -LE hwd`):

Action	Result
xpm run install-all	rc 0
xpm run test-native-cmake-sys	16/16 in debug and release
xpm run test-cortex-cmake	3/3 in each of the 8 configurations
xpm run test-smp-cmake	aarch32 and aarch64 rpi-zero-2w / rpi3b 15/15, 2xcortex-m33 4/4, pico2-1cpu 4/4, cortexm-pico2 6/6, cortexm-pico2-rp2350b-psram 3/3 — debug and release, rc 0

In the first pass of `test-smp-cmake`, `aarch32-rpi-zero-2w-smp-pipeline-test-qemu` (release) stopped printing after 26 s of guest time and was killed at its 1000 s timeout. Run alone it then passed 5 times in 5 (35 s each), and the complete second pass of `test-smp-cmake` was green. It is recorded as an intermittent stall of the SMP pipeline test under QEMU (`raspi3b`, 4 cores), to be investigated separately; it is not a defect of the migration series.

22. Complete Pull-Request and Commit List (all repositories)

Every commit of every repository is one pull request. A PR's head is the local branch `pr/<Commit-ID>`, which points at that commit; its base is the previous PR's branch in the same repository (the first PR of a repository targets `xpack-development`). After a PR is merged, the next one is retargeted to `xpack-development` (§12.5). The branches exist locally in each `/tmp/<repository>` clone and are pushed to `github.com/dan-maio/<repository>` only when the PRs are opened.

22.1 Legend and totals

- **PR:** position on `smp`, counted from the base (§20.1); this is the order in which the repository's PRs are opened and merged.
- **Head branch (Commit-ID):** `pr/<ID>`, where `<ID>` is the ID in the commit trailer `Commit-ID: <ID>` (`xpack-dev-smp.md Part II`).
- **Base branch:** the branch the PR targets.
- **Category:** `correction` (fix), `new code` (feat), `test`, `build`, `docs`, `chore`, `tool`.
- **Stage / theme:** the Part II stage (§13.1), followed by the R6 group (§21.2) when the commit belongs to one.

Repository	Corrections	New code	Tests	Build	Docs	Chore / tool	PRs
micro-os-plus-iii (migration)	31	18	32	3	39	2	125
micro-os-plus-iii (documentati on updates, §22.4)	0	0	0	0	34	0	34
micro-os-plus-iii-posix- arch	0	5	15	2	0	2	24
micro-os-plus-iii- cortexm	1	9	72	2	1	1	86
micro-os-plus-iii- aarch32	0	10	52	1	0	1	64
micro-os-plus-iii- aarch64	0	9	32	1	0	1	43
micro-os-plus-iii- devices	0	0	0	0	1	0	1
Total	32	51	203	9	75	7	377

22.2 Cross-repository merge order

The PRs of one repository merge in their own order (22.3). Across repositories, R6 (provider before consumer) adds these constraints:

Before	Merge	Why
—	S0 in every port: posix-arch PR 1–3, cortexm PR 1–3, aarch32 PR 1–3, aarch64 PR 1–2; then devices PR 1	The drivers and SoC support exist in the ports before the devices repository is retired.
kernel K28a	G-hrclock together: kernel K28 + cortexm C04, C04b + posix-arch P04	The kernel calls <code>clock_highres::has_hardware_counter()</code> , which each port defines.
kernel K39b	posix-arch P05, P06	The host SMP port relies on the kernel's tolerant <code>port_cpu_id()</code> declarations.
kernel K30–K39b, K40, K41, K43a–d	G-harness together: kernel K42, K44, K45, K46-m0..m7f + posix-arch P20 + cortexm C20a, C20b	The kernel harness switches to the port test builders; neither side builds alone.
cortexm C09 / C10 and its board PR	kernel K50 (<code>2xcortex-m33</code>), K51 (<code>pico2-1cpu</code>), K52–K57 (one per cortexm board)	A kernel platform needs its port core and board.
aarch32 A32-20, A32-21 (and A32-R3, A32-LL)	kernel K58, K59, K62	idem for AArch32.
aarch64 A64-20, A64-21 (and A64-R3)	kernel K60, K61	idem for AArch64.
the board PR of a port	that board's test PRs (<code>pr/<board-ID>-<test></code>)	One test directory = one <code>ctest</code> entry on an existing platform.

22.3 Pull requests per repository

micro-os-plus-iii -- 159 pull requests (github.com/dan-maio/micro-os-plus-iii)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/K01	xpack-development	b364e689	correction	S1 single-core	fix(posix): ISO C conformant DIR in dirent.h
2	pr/K02	pr/K01	52a7f499	correction	S1 single-core	fix(utils): list iterators call node_- >next()/ prev()
3	pr/K03	pr/K02	2f8c482e	correction	S1 single-core	fix(posix- io): match newlib's syscall return types in the weak aliases
4	pr/K04	pr/K03	bcb5737e	correction	S1 single-core	fix(rtos): emit this_thread: :suspend() out of line
5	pr/K06	pr/K04	ef9bb2ce	correction	S1 single-core	fix(memory): block_pool asserts on a null result, not on a valid one
6	pr/K07	pr/K06	a6d6c475	correction	S1 single-core	fix(estd): polymorphic_ allocator copy keeps its memory resource
7	pr/K08	pr/K07	51bdad10	new code	S1 single-core	feat(memory): usable size for the RTOS allocators, overflow- checked calloc()
8	pr/K09	pr/K08	1c589065	new code	S1 single-core	feat(libcpp): C++17 aligned operator new/delete
9	pr/K10	pr/K09	de47254b	correction	S1 single-core	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						<code>fix(libc++): error categories with static storage</code>
10	<code>pr/K11</code>	<code>pr/K10</code>	<code>c8aa3024</code>	correction	S1 single-core	<code>fix(libc++): steady_clock ::now() without 64- bit overflow</code>
11	<code>pr/K12</code>	<code>pr/K11</code>	<code>0c11e258</code>	correction	S1 single-core	<code>fix(rtos): C API timers default to one-shot</code>
12	<code>pr/K13</code>	<code>pr/K12</code>	<code>ebfa07b3</code>	correction	S1 single-core	<code>fix(rtos): C API deletes mutexes and semaphores through their concrete type</code>
13	<code>pr/K14</code>	<code>pr/K13</code>	<code>b4d6fddf</code>	correction	S1 single-core	<code>fix(rtos): widen millisec before scaling in the CMSIS- RTOS API</code>
14	<code>pr/K15</code>	<code>pr/K14</code>	<code>8ed1c7b0</code>	correction	S1 single-core	<code>fix(posix- io): serialize the file descriptor table</code>
15	<code>pr/K16</code>	<code>pr/K15</code>	<code>fa232ed7</code>	correction	S1 single-core	<code>fix(posix- io): lock the deferred file-system and socket lists</code>
16	<code>pr/K17</code>	<code>pr/K16</code>	<code>01b50480</code>	correction	S1 single-core	<code>fix(posix- io): block_device size queries tested the wrong condition</code>

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
17	pr/K18	pr/K17	deeabab3	new code	S1 single-core	<code>feat(arm): semihosting trap for ARMv8-M, and an optional HLT trap</code>
18	pr/K19	pr/K18	b85b378a	new code	S1 single-core	<code>feat(startup): exception handlers and fault dump for ARMv8-M Mainline</code>
19	pr/K20a	pr/K19	182b5f41	correction	S1 single-core	<code>fix(semihosting): fstat() keeps a file type the caller already set</code>
20	pr/K20b	pr/K20a	98eaf96f	new code	S1 single-core	<code>feat(semihosting): weak board console mirror for stdout/ stderr</code>
21	pr/K21a	pr/K20b	236b2d1a	correction	S1 single-core	<code>fix(rtos): errno in handler mode does not assert</code>
22	pr/K21	pr/K21a	8ea2f2f6	correction	S1 single-core	<code>fix(rtos): run timer callbacks outside the critical section</code>
23	pr/K23a	pr/K21	267b955b	correction	S1 single-core	<code>fix(rtos): refuse a priority- ceiling lock before taking ownership</code>
24	pr/K23b	pr/K23a	f5861fc2	correction	S1 single-core	<code>fix(rtos): priority inheritance keeps the highest</code>

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						boost among waiters
25	pr/K23c	pr/K23b	e2b3709c	correction	S1 single-core	fix(rtos): mutex lock re-checks the owner after boosting it
26	pr/K24a	pr/K23c	5390f31d	new code	S1 single-core	feat(rtos): thread state destroying, claimed by exactly one destroyer
27	pr/K24b	pr/K24a	8fcfbc41	correction	S1 single-core	fix(rtos): resume() re- enqueues only a suspended or initializing thread
28	pr/K24c	pr/K24b	61756303	correction	S1 single-core	fix(rtos): priority changes test and relink under one lock
29	pr/K24d	pr/K24c	94dce976	new code	S1 single-core	feat(rtos): implement thread::deta ch()
30	pr/K24e	pr/K24d	af3dfd7b	correction	S1 single-core	fix(rtos): join() and thread destruction without a lost wake- up
31	pr/K25a	pr/K24e	325a75cc	correction	S1 single-core	fix(estd): condition_va riable::wait _for() trusts the kernel's timeout verdict
32	pr/K25b	pr/K25a	102a9136	correction	S1 single-core	fix(rtos): condition variable without lost

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						signals, bound to its clock
33	pr/K26a	pr/K25b	e57aff39	correction	S1 single-core	<code>fix(estd):</code> std::thread keeps its own pointer to the bound function object
34	pr/K26b	pr/K26a	8742e5f4	correction	S1 single-core	<code>fix(estd):</code> std::thread: :join() waits for the thread before freeing it
35	pr/K27	pr/K26b	72af88d8	correction	S1 single-core	<code>fix(rtos):</code> message queue reschedules after waking a peer
36	pr/K28a	pr/K27	135edeb3	correction	S1 single-core	<code>fix(rtos):</code> <cmsis-plus/ rtos/os- decls.h> includes the C declarations again
37	pr/K28	pr/K28a	b474cf72	new code	S1 single-core G-hrclock	<code>feat(rtos):</code> the high- resolution clock reads the port's hardware counter
38	pr/K29a	pr/K28	4804b93c	test	S1 single-core	<code>test(rtos- apis):</code> a 512 KiB RTOS arena
39	pr/K29b	pr/K29a	675a698e	test	S1 single-core	<code>test(rtos- apis):</code> OS_EXCLUDE_R TOS_APIS_FAT FS leaves the FatFs leg out

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
40	pr/K29c	pr/K29b	f242e451	correction	S1 single-core	<code>fix(test):</code> enable the FPU when the compiler targets one
41	pr/K29d	pr/K29c	9cb742d0	docs	S1 single-core	<code>docs(test):</code> the MPS2 AN385/AN386 flash comment says 128M
42	pr/K30	pr/K29d	6f51da7a	new code	S2 SMP core	<code>feat(port):</code> smp-common -- the SMP declarations a thin port shares
43	pr/K31	pr/K30	f1894152	new code	S2 SMP core	<code>feat(rtos):</code> one current thread per CPU
44	pr/K32	pr/K31	604321a6	new code	S2 SMP core	<code>feat(rtos):</code> thread CPU affinity
45	pr/K33	pr/K32	ccfc03ec	new code	S2 SMP core	<code>feat(rtos):</code> SMP thread picker with affinity and the stack- pointer claim
46	pr/K34	pr/K33	3764d613	new code	S2 SMP core	<code>feat(rtos):</code> wake an eligible CPU with an IPI when a thread is resumed elsewhere
47	pr/K35	pr/K34	ce44e5b8	new code	S2 SMP core	<code>feat(rtos):</code> <code>join()</code> returns only once the thread is off every CPU
48	pr/K36	pr/K35	1a811b99	new code	S2 SMP core	<code>feat(rtos):</code> <code>kill()</code> waits until the

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						thread is off every CPU and unclaimed
49	pr/K37	pr/K36	a5194f6e	new code	S2 SMP core	<code>feat(rtos):</code> the idle reaper leaves a thread that is still live on another CPU
50	pr/K38	pr/K37	bcab09e7	new code	S2 SMP core	<code>feat(rtos):</code> pin the main thread to CPU 0
51	pr/K39	pr/K38	b4e819eb	new code	S2 SMP core	<code>feat(rtos):</code> CMSIS-RTOS v1 threads run on CPU 0
52	pr/K39b	pr/K39	12882e19	correction	S2 SMP core	<code>fix(rtos):</code> the kernel's <code>port_cpu_id(</code> <code>)</code> declarations tolerate the port's own
53	pr/K40	pr/K39b	1f6d219f	build	S4a harness	<code>build(cmake)</code> : toolchain profiles for arm-none- eabi, aarch64- none-elf and the host
54	pr/K41	pr/K40	68155aee	build	S4a harness	<code>build(cmake)</code> : <code>uos_add_app(</code> <code>) -- one</code> call declares an application against a port
55	pr/K43a	pr/K41	0b6d9339	test	S4a harness	<code>test:</code> the verdict contract every board shares

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						(hw_result, board contract)
56	pr/K43b	pr/K43a	a569c214	test	S4a harness	test: the host runner for the port test application s
57	pr/K43c	pr/K43b	4db26232	test	S4a harness	test: the QEMU runner for the port test application s
58	pr/K43d	pr/K43c	0092d31d	test	S4a harness	test: the hardware runner and the OpenOCD configs that need no power cycle
59	pr/K42	pr/K43d	a530fac4	build	S4a harness · G-harness	build(cmake) : modular kernel targets
60	pr/K44	pr/K42	2f21e8ca	test	S4a harness · G-harness	test(cmake): the library under test is the architecture port, found as a sibling
61	pr/K45	pr/K44	5fef63ae	test	S4a harness · G-harness	test(native) : run posix- arch's own test application s
62	pr/K46-m0	pr/K45	62d55a6e	test	S4a harness · G-harness	test(qemu- cortex-m0): build against the cortexm port's generic QEMU core
63	pr/K46-m3	pr/K46-m0	acc3e284	test		test(qemu- cortex-m3):

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
					S4a harness · G-harness	build against the cortexm port's generic QEMU core
64	pr/K46-m4f	pr/K46-m3	b3322cc3	test	S4a harness · G-harness	test(qemu- cortex-m4f): build against the cortexm port's generic QEMU core
65	pr/K46-m7f	pr/K46-m4f	d0edbce3	test	S4a harness · G-harness	test(qemu- cortex-m7f): build against the cortexm port's generic QEMU core
66	pr/K47	pr/K46-m7f	c91f3b6b	test	S4a harness	test(qemu): MPS2 AN505/ AN521 memory maps and ARMv8-M support in the QEMU device
67	pr/K49	pr/K47	9a194e45	test	S4b test suite	test: fp- switch -- the FPU registers survive a preemptive switch and a core move
68	pr/K50	pr/K49	8efc3ecb	test	S4c platform	test(2xcorte x-m33): two Cortex-M33 cores on QEMU mps2- an521
69	pr/K51	pr/K50	85c38290	test	S4c platform	test(pico2-1 cpu): the RP2350's Cortex-M33 code at one CPU on QEMU

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
70	pr/K52	pr/K51	5af5b8d7	test	S4c platform	<code>test(cortexm-pico2):</code> the Pico 2 board's own tests, on QEMU and on the board
71	pr/K53	pr/K52	41ee57a4	test	S4c platform	<code>test(cortexm-pico2-rp2350b-psram):</code> the RP2350B PSRAM board's tests
72	pr/K54	pr/K53	e5a581a8	test	S4c platform	<code>test(cortexm-pico2-pizero):</code> the pico2-pizero board's tests, hardware only
73	pr/K55	pr/K54	76fd013c	test	S4c platform	<code>test(cortexm-nucleof411):</code> the NUCLEO-F411RE board's tests under the SMP scheduler at one CPU
74	pr/K56	pr/K55	c0772ab8	test	S4c platform	<code>test(cortexm-weactf411):</code> the WeAct F411 board's tests
75	pr/K57	pr/K56	084b6528	test	S4c platform	<code>test(cortexm-weactf412):</code> the WeAct F412 board's tests
76	pr/K58	pr/K57	93faa7da	test	S4c platform	<code>test(aarch32-rpi-zero-2w):</code> the Zero 2 W's AArch32 tests on

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						QEMU raspi3b
77	pr/K59	pr/K58	8f3a1051	test	S4c platform	test(aarch32-rpi3b): the Pi 3 B's AArch32 tests on QEMU raspi3b
78	pr/K62	pr/K59	9ee32a05	test	S4c platform	test(aarch32-luckfox-lyra): build the Luckfox Lyra's tests
79	pr/K60	pr/K62	55841a50	test	S4c platform	test(aarch64-rpi-zero-2w): the Zero 2 W's AArch64 tests on QEMU raspi3b
80	pr/K61	pr/K60	eb23b89a	test	S4c platform	test(aarch64-rpi3b): the Pi 3 B's AArch64 tests on QEMU raspi3b
81	pr/K39c	pr/K61	c329210b	test	S2 SMP core	test(validator): the host build uses glibc's ucontext, not libucontext
82	pr/K48-nucleo-f411re	pr/K39c	a62e6992	test	S4a harness	test(nucleo-f411re): label the board tests hwd and link the kernel's modular targets
83	pr/K48-nucleo-f767zi	pr/K48-nucleo-f411re	0f3d65ad	test	S4a harness	test(nucleo-f767zi): label the board tests hwd and link

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						the kernel's modular targets
84	pr/K48-nucleo-h743zi	pr/K48-nucleo-f767zi	cfa25b4c	test	S4a harness	test(nucleo-h743zi): label the board tests hwd and link the kernel's modular targets
85	pr/K48-raspberrypi-pico	pr/K48-nucleo-h743zi	87be7f54	test	S4a harness	test(raspberrypi-pico): label the board tests hwd and link the kernel's modular targets
86	pr/K901	pr/K48-raspberrypi-pico	eb6fb4d9	docs	S5 docs/chore	docs: aarch32-second-board
87	pr/K902	pr/K901	b8c45608	docs	S5 docs/chore	docs: building-aarch32-aarch64
88	pr/K903	pr/K902	5f899167	docs	S5 docs/chore	docs: cortexm-port
89	pr/K904	pr/K903	8800c1c7	docs	S5 docs/chore	docs: posix-arch-port
90	pr/K905	pr/K904	c3f5b83a	docs	S5 docs/chore	docs: smp-integration/ agy-review
91	pr/K906	pr/K905	b2027337	docs	S5 docs/chore	docs: smp-integration/ DeepSeek-review
92	pr/K907	pr/K906	c98fcc93	docs	S5 docs/chore	docs: smp-integration/ diagrams
93	pr/K908	pr/K907	8ddcb6c6	docs	S5 docs/chore	docs: smp-integration/ files-modif-by-step
94	pr/K909	pr/K908	b9695be3	docs	S5 docs/chore	docs: smp-integration/

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						GITHUB- PROGRESSIVE- PR-GUIDE
95	pr/K910	pr/K909	ac125125	docs	S5 docs/chore	docs: smp- integration/ Implementati on-SMP- Integration
96	pr/K911	pr/K910	75e7e1c3	docs	S5 docs/chore	docs: smp- integration/ micro-os- plus-iii- project- unification
97	pr/K912	pr/K911	0e26f6e2	docs	S5 docs/chore	docs: smp- integration/ MICRO-OS- PLUS-SMP-VS- SINGLECORE- ANALYSIS
98	pr/K913	pr/K912	371d71b1	docs	S5 docs/chore	docs: smp- integration/ new- modification s
99	pr/K914	pr/K913	8930f3c7	docs	S5 docs/chore	docs: smp- integration/ pull- request
100	pr/K915	pr/K914	6658ffb3	docs	S5 docs/chore	docs: smp- integration/ smp- constructio n
101	pr/K916	pr/K915	28e2fb7d	docs	S5 docs/chore	docs: smp- integration/ SMP- UPSTREAM- INTEGRATION- PLAN
102	pr/K917	pr/K916	a840dec4	docs	S5 docs/chore	docs: smp- integration/ xpack-dev- smp
103	pr/K918	pr/K917	8fdb54ca	docs	S5 docs/chore	docs: specs
104	pr/K919	pr/K918	32f6644e	docs	S5 docs/chore	docs: STATUS
105	pr/K920	pr/K919	5e847731	docs	S5 docs/chore	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						docs: tests/AARCH32-RPI-ZERO-2W-TESTS
106	pr/K921	pr/K920	d925dc2c	docs	S5 docs/chore	docs: tests/HARNESS-BOARD-TEST-CHEATSHEET
107	pr/K922	pr/K921	bc6c523b	docs	S5 docs/chore	docs: tests/HARNESS-TESTS-PARADIGM
108	pr/K923	pr/K922	8c4fbd55	docs	S5 docs/chore	docs: tests-in-aarch32-aarch64
109	pr/K924	pr/K923	646cb078	docs	S5 docs/chore	docs: test-smpl
110	pr/K925	pr/K924	a136e8bd	docs	S5 docs/chore	docs: tests/README
111	pr/K926	pr/K925	20bf4b20	docs	S5 docs/chore	docs: tests/README-DEVELOPER
112	pr/K927	pr/K926	9b8d0fe9	docs	S5 docs/chore	docs: tests/README-MAINTAINER
113	pr/K928	pr/K927	d53e34c7	docs	S5 docs/chore	docs: tests/STEPS
114	pr/K929	pr/K928	3a0b2153	docs	S5 docs/chore	docs: tests/test-framework
115	pr/K930	pr/K929	4815480b	docs	S5 docs/chore	docs: tests/TESTS-CATALOG
116	pr/K931	pr/K930	d142d5a1	docs	S5 docs/chore	docs: tests/TESTS-DEVELOPER-GUIDE
117	pr/K932	pr/K931	c72b7500	docs	S5 docs/chore	docs: tests/TESTS-XPACK-SYSTEM
118	pr/K933	pr/K932	1deca033	docs	S5 docs/chore	docs: tests/WORK-SMP-AARCH32-AARCH64-HARNESS-GUIDE

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
119	pr/K934	pr/K933	72f5d7a1	docs	S5 docs/chore	<code>docs:</code> <code>upstream-</code> <code>CHANGELOG</code>
120	pr/K935	pr/K934	6e0c1276	docs	S5 docs/chore	<code>docs:</code> <code>upstream-</code> <code>package</code>
121	pr/K980	pr/K935	675652cb	docs	S5 docs/chore	<code>docs:</code> <code>README-</code> <code>DEVELOPER</code> updated from the reference <code>smp branch</code>
122	pr/K981	pr/K980	f6337d83	docs	S5 docs/chore	<code>docs(test):</code> <code>tests/README</code> for the 22 platforms, <code>TO-CHECK</code> marked as <code>history</code>
123	pr/K982	pr/K981	0301394d	chore	S5 docs/chore	<code>chore:</code> a <code>.clang-</code> <code>format</code> for the kernel sources
124	pr/K983	pr/K982	c7541e17	chore	S5 docs/chore	<code>chore:</code> ignore every <code>build*</code> <code>directory</code> and the <code>build</code> products
125	pr/K984	pr/K983	ba3817b5	docs	S5 docs/chore	<code>docs(smp):</code> <code>xpack-dev-</code> <code>smp.md</code> Part II -- the commit-only migration procedure
126	pr/K985	pr/K984	6fb211a9	docs	S5 docs/chore	<code>docs(smp):</code> <code>xpack-dev-</code> <code>smp.md</code> Part III -- execution record, step summary, commit list
127	pr/K986- STATUS	pr/K985	e2a90a42	docs	S5 docs/chore	<code>docs:</code> <code>STATUS</code> for the

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						commit-only migration of 2026-10-06
128	pr/K986-aarch32-second-board	pr/K986-STATUS	7bfb05be	docs	S5 docs/chore	docs: aarch32-second-board -- the RK3506 SD driver is in the port
129	pr/K986-building-aarch32-aarch64	pr/K986-aarch32-second-board	a256bafe	docs	S5 docs/chore	docs: building-aarch32-aarch64 without the devices repository
130	pr/K986-posix-arch-port	pr/K986-building-aarch32-aarch64	6c6a136a	docs	S5 docs/chore	docs: posix-arch-port -- drivers and SD back-end are in the port
131	pr/K986-cortexm-port	pr/K986-posix-arch-port	a9c1ad47	docs	S5 docs/chore	docs: cortexm-port -- the SoC directories and the verified results
132	pr/K986-test-smpl	pr/K986-cortexm-port	c273abb1	docs	S5 docs/chore	docs: test-smpl lists the verdict contract and the no- power-cycle configs
133	pr/K986-tests-in-aarch32-aarch64	pr/K986-test-smpl	0ff357af	docs	S5 docs/chore	docs: tests-in-aarch32-aarch64 shows drivers/ and soc/ in the port
134	pr/K986-WORK-SMP-HARNESS-GUIDE	pr/K986-tests-in-aarch32-aarch64	359b2b73	docs	S5 docs/chore	docs(tests): WORK-SMP harness guide -- no devices repository

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
135	<code>pr/K986-TESTS-XPACK-SYSTEM</code>	<code>pr/K986-WORK-SMP-HARNESS-GUIDE</code>	<code>982522d7</code>	docs	S5 docs/chore	<code>docs(tests): TESTS-XPACK-SYSTEM -- the four ports, devices dissolved</code>
136	<code>pr/K986-HARNESS-TESTS-PARADIGM</code>	<code>pr/K986-TESTS-XPACK-SYSTEM</code>	<code>a8983cd6</code>	docs	S5 docs/chore	<code>docs(tests): HARNESS-TESTS-PARADIGM -- no devices repository</code>
137	<code>pr/K986-HARNESS-BOARD-TEST-CHEATSHEET</code>	<code>pr/K986-HARNESS-TESTS-PARADIGM</code>	<code>ef3d8096</code>	docs	S5 docs/chore	<code>docs(tests): HARNESS-BOARD-TEST-CHEATSHEET -- no devices repository</code>
138	<code>pr/K986-AARCH32-RPI-ZERO-2W-TESTS</code>	<code>pr/K986-HARNESS-BOARD-TEST-CHEATSHEET</code>	<code>997146ca</code>	docs	S5 docs/chore	<code>docs(tests): AARCH32-RPI-ZERO-2W-TESTS -- drivers are in the port</code>
139	<code>pr/K986-STEPS</code>	<code>pr/K986-AARCH32-RPI-ZERO-2W-TESTS</code>	<code>335b3416</code>	docs	S5 docs/chore	<code>docs(tests): STEPS -- a port finds only the kernel as a sibling</code>
140	<code>pr/K986-TESTS-DEVELOPER-GUIDE</code>	<code>pr/K986-STEPS</code>	<code>3ee23fe2</code>	docs	S5 docs/chore	<code>docs(tests): TESTS-DEVELOPER-GUIDE -- the workspace without devices</code>
141	<code>pr/K986-TESTS-CATALOG</code>	<code>pr/K986-TESTS-DEVELOPER-GUIDE</code>	<code>63f2b822</code>	docs	S5 docs/chore	<code>docs(tests): TESTS-CATALOG -- four ports, and the last verified run</code>
142	<code>pr/K986-tests-README-DEVELOPER</code>	<code>pr/K986-TESTS-CATALOG</code>	<code>d504cbc4</code>	docs	S5 docs/chore	<code>docs(tests): README-DEVELOPER -- the current</code>

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						clone layout
143	pr/K986- tests- README- MAINTAINER	pr/K986- tests- README- DEVELOPER	0ddf9763	docs	S5 docs/chore	docs(tests): README- MAINTAINER -- the current clone layout
144	pr/K986- test- framework	pr/K986- tests- README- MAINTAINER	b60858d1	docs	S5 docs/chore	docs(tests): test- framework points to the smp side
145	pr/K986-agy- review	pr/K986- test- framework	8c3477d2	docs	S5 docs/chore	docs(smp): agy-review -- status note, superseded by xpack- dev-smp.md
146	pr/K986- DeepSeek- review	pr/K986-agy- review	60e4d0ef	docs	S5 docs/chore	docs(smp): DeepSeek- review -- status note, superseded by xpack- dev-smp.md
147	pr/K986- files-modif- by-step	pr/K986- DeepSeek- review	1da601cd	docs	S5 docs/chore	docs(smp): files-modif- by-step -- status note, superseded by xpack- dev-smp.md
148	pr/K986- GITHUB- PROGRESSIVE- PR-GUIDE	pr/K986- files-modif- by-step	66ef2fa5	docs	S5 docs/chore	docs(smp): GITHUB- PROGRESSIVE- PR-GUIDE -- status note, superseded by xpack- dev-smp.md
149	pr/K986- Implementati on-SMP- Integration	pr/K986- GITHUB- PROGRESSIVE- PR-GUIDE	440716ba	docs	S5 docs/chore	docs(smp): Implementati on-SMP- Integration -- status note,

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						superseded by xpack- dev-smp.md
150	pr/K986- micro-os- plus-iii- project- unification	pr/K986- Implementati on-SMP- Integration	c6c7eebe	docs	S5 docs/chore	docs(smp): project- unification -- status note
151	pr/K986- MICRO-OS- PLUS-SMP-VS- SINGLECORE- ANALYSIS	pr/K986- micro-os- plus-iii- project- unification	df0f0ed9	docs	S5 docs/chore	docs(smp): SMP-VS- SINGLECORE- ANALYSIS -- status note
152	pr/K986-new- modification s	pr/K986- MICRO-OS- PLUS-SMP-VS- SINGLECORE- ANALYSIS	2be02079	docs	S5 docs/chore	docs(smp): new- modification s -- status note
153	pr/K986- pull- request	pr/K986-new- modification s	0e1416d3	docs	S5 docs/chore	docs(smp): pull-request -- status note, superseded by xpack- dev-smp.md
154	pr/K986-smp- constructio n	pr/K986- pull- request	02e62612	docs	S5 docs/chore	docs(smp): smp- construction -- status note
155	pr/K986-SMP- UPSTREAM- INTEGRATION- PLAN	pr/K986-smp- constructio n	21d772ad	docs	S5 docs/chore	docs(smp): SMP- UPSTREAM- INTEGRATION- PLAN -- status note, superseded by xpack- dev-smp.md
156	pr/ K986-2026-09 -20-micro- os-plus-iii- smp- unification- design	pr/K986-SMP- UPSTREAM- INTEGRATION- PLAN	039c42c6	docs	S5 docs/chore	docs(specs): smp- unification- design -- status note
157	pr/K987	pr/ K986-2026-09 -20-micro- os-plus-iii-	646a4904	docs	S5 docs/chore	docs(smp): xpack-dev- smp.md lists the

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
		smp- unification- design				documentatio n updates
158	pr/K988	pr/K987	ec40386b	docs	S5 docs/chore	docs(smp): xpack-dev- smp.md -- the pull requests of every repository
159	pr/K989	pr/K988	(this commit)	docs	S5 docs/chore	docs(smp): xpack-dev- smp.md -- the verification replay from golden

micro-os-plus-iii-posix-arch -- 24 pull requests (github.com/dan-maio/micro-os-plus-iii-posix-arch)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	<code>pr/P01</code>	<code>xpack-development</code>	<code>26a120bf</code>	new code	S0 devices	<code>feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers</code>
2	<code>pr/P02</code>	<code>pr/P01</code>	<code>b8e8b4ad</code>	new code	S0 devices	<code>feat(soc): import the native host-file SD backend</code>
3	<code>pr/P03</code>	<code>pr/P02</code>	<code>9c5914ed</code>	build	S0 devices	<code>build(cmake): define the dissolved devices targets locally</code>
4	<code>pr/P04</code>	<code>pr/P03</code>	<code>5a1a255a</code>	new code	S1 hrclock · G-hrclock	<code>feat(port): clock_highres reads CLOCK_MONOTONIC</code>
5	<code>pr/P05</code>	<code>pr/P04</code>	<code>9be889da</code>	new code	S2 host SMP	<code>feat(port): the host is a multi-core machine -- one host thread per CPU</code>
6	<code>pr/P06</code>	<code>pr/P05</code>	<code>22f54df7</code>	new code	S2 host SMP	<code>feat(port): the board verdict contract on the host</code>
7	<code>pr/P20</code>	<code>pr/P06</code>	<code>e4f6c62f</code>	build	S4 harness · G-harness	<code>build(cmake): the port selects its board, and builds its own tests</code>
8	<code>pr/P21</code>	<code>pr/P20</code>	<code>305de44c</code>	test	S4 test	<code>test(native): rtos-apis at OS_NCPU = 1</code>
9	<code>pr/P22</code>	<code>pr/P21</code>	<code>e080ea6a</code>	test	S4 test	<code>test(native): mutex-stress</code>
10	<code>pr/P23</code>	<code>pr/P22</code>	<code>76c72864</code>	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						test(native) : flatfs- test -- append to a file created empty
11	pr/P24	pr/P23	e648c634	test	S4 test	test(native) : mutex- ceiling-test -- a refused protect lock leaves nothing behind
12	pr/P25	pr/P24	f0820dd5	test	S4 test	test(native) : smp_test0 -- bring-up on one active CPU
13	pr/P26	pr/P25	c249804c	test	S4 test	test(native) : smp_test1 -- semaphore ping-pong across CPUs
14	pr/P27	pr/P26	4f1eee6b	test	S4 test	test(native) : smp_test2 -- kernel lock coherency on every CPU
15	pr/P28	pr/P27	c26ff0fe	test	S4 test	test(native) : smp_test3 -- message queue producer/ consumer across CPUs
16	pr/P29	pr/P28	b67adcae	test	S4 test	test(native) : smp_test4 -- load balancing without affinity
17	pr/P30	pr/P29	75a487c7	test	S4 test	test(native) : smp-num- test -- concurrent workers and console I/O

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
18	pr/P31	pr/P30	9fe414ca	test	S4 test	<code>test(native)</code> <code>: smp-mat-</code> <code>test --</code> <code>parallel</code> <code>block solver</code> <code>(N=500,</code> <code>B=50)</code>
19	pr/P32	pr/P31	e60f2f2c	test	S4 test	<code>test(native)</code> <code>: smp-</code> <code>pipeline-</code> <code>test --</code> <code>multi-stage</code> <code>pipeline</code> <code>across CPUs</code>
20	pr/P33	pr/P32	e614e825	test	S4 test	<code>test(native)</code> <code>: smp-pro-</code> <code>cons-test --</code> <code>producers</code> <code>and</code> <code>consumers</code> <code>over every</code> <code>kernel</code> <code>object</code>
21	pr/P34	pr/P33	c4f5f652	test	S4 test	<code>test(native)</code> <code>: smp-mutex-</code> <code>stress --</code> <code>the SMP leg</code> <code>of mutex-</code> <code>stress, with</code> <code>per-core</code> <code>checks</code>
22	pr/P35	pr/P34	6aec9d4c	test	S4 test	<code>test(native)</code> <code>: smp-rtos-</code> <code>apis -- the</code> <code>SMP leg of</code> <code>rtos-apis</code>
23	pr/P14	pr/P35	e156e975	tool	tool	<code>tools: a</code> <code>probe</code> <code>showing why</code> <code>TSan fiber</code> <code>annotations</code> <code>cannot work</code> <code>here</code>
24	pr/P36	pr/P14	93c42239	chore	chore	<code>chore:</code> <code>ignore every</code> <code>build*</code> <code>directory</code> <code>and the</code> <code>build</code> <code>products</code>

micro-os-plus-iii-cortexm -- 86 pull requests (github.com/dan-maio/micro-os-plus-iii-cortexm)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	<code>pr/C01</code>	<code>xpack-development</code>	<code>b2bdd12c</code>	new code	S0 devices	<code>feat(soc):</code> <code>import the</code> <code>STM32F4xx</code> <code>SoC support</code>
2	<code>pr/C02</code>	<code>pr/C01</code>	<code>2310c355</code>	new code	S0 devices	<code>feat(soc):</code> <code>import the</code> <code>RP2350 SoC</code> <code>support</code>
3	<code>pr/C03</code>	<code>pr/C02</code>	<code>f64c09a4</code>	build	S0 devices	<code>build(cmake)</code> <code>: define the</code> <code>dissolved</code> <code>SoC targets</code> <code>locally</code>
4	<code>pr/C04</code>	<code>pr/C03</code>	<code>736010e8</code>	new code	S1 hrclock · G-hrclock	<code>feat(port):</code> <code>clock_highre</code> <code>s declares</code> <code>it has no</code> <code>hardware</code> <code>counter</code>
5	<code>pr/C04b</code>	<code>pr/C04</code>	<code>31b04065</code>	correction	S1 hrclock	<code>fix(port):</code> <code>cycles_since</code> <code>_tick()</code> <code>reads the</code> <code>SysTick</code> <code>pending flag</code> <code>from ICSR</code>
6	<code>pr/C05</code>	<code>pr/C04b</code>	<code>dfbfda56</code>	new code	S2 Cortex-M SMP	<code>feat(port):</code> <code>the SMP port</code> <code>contract on</code> <code>the generic</code> <code>Cortex-M</code> <code>core</code>
7	<code>pr/C06</code>	<code>pr/C05</code>	<code>cc6412ab</code>	new code	S2 Cortex-M SMP	<code>feat(port):</code> <code>the generic</code> <code>Cortex-M</code> <code>core runs</code> <code>the SMP</code> <code>scheduler on</code> <code>one CPU</code>
8	<code>pr/C07</code>	<code>pr/C06</code>	<code>c9b5c630</code>	new code	S2 Cortex-M SMP	<code>feat(libc):</code> <code>a</code> <code>_getentropy(</code> <code>) stub for</code> <code>the QEMU</code> <code>images</code>
9	<code>pr/C08</code>	<code>pr/C07</code>	<code>597bdda8</code>	new code	S2 Cortex-M SMP	<code>feat:</code> <code>_Exit()</code> <code>reports the</code> <code>test verdict</code>

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						through semihosting
10	pr/C09	pr/C08	ba9ef5ee	new code	S3 new silicon	feat(port): a generic Cortex-M33 (SSE-200) dual-core port for QEMU
11	pr/C10	pr/C09	0071da04	new code	S3 new silicon	feat(port): the RP2350 dual-core port
12	pr/C20a	pr/C10	591b070e	test	S4 board/ harness	test(nucleof 411): the NUCLEO- F411RE board
13	pr/C20b	pr/C20a	3629ead6	build	S4 board/ harness · G- harness	build(cmake) : the port selects its board, builds its tests, and exports the QEMU cores
14	pr/C21	pr/C20b	2187bd34	test	S4 board/ harness	test(pico2): the Raspberry Pi Pico 2 board (RP2350, two Cortex-M33)
15	pr/C21b	pr/C21	fbd44d24	test	S4 board/ harness	test(pico2): vendor the TinyUSB device stack for the RP2350
16	pr/C21c	pr/C21b	e82f3f7e	test	S4 board/ harness	test(pico2): a minimal pico-sdk shim for TinyUSB
17	pr/C21d	pr/C21c	9e62cb6b	test	S4 board/ harness	test(pico2): CDC-ACM and HID device configuratio ns
18		pr/C21d	13e72795	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
	pr/C22-rtos-apis					test(pico2): rtos-apis
19	pr/C22-cmsis-os-validator-ram	pr/C22-rtos-apis	094d7daf	test	S4 test	test(pico2): cmsis-os-validator-ram
20	pr/C22-cmsis-os-validator	pr/C22-cmsis-os-validator-ram	e58e7663	test	S4 test	test(pico2): cmsis-os-validator
21	pr/C22-exc-test	pr/C22-cmsis-os-validator	49e47089	test	S4 test	test(pico2): exc-test
22	pr/C22-fp-switch	pr/C22-exc-test	3cbb8aff	test	S4 test	test(pico2): fp-switch
23	pr/C22-mutex-stress-ram	pr/C22-fp-switch	88bd02ad	test	S4 test	test(pico2): mutex-stress-ram
24	pr/C22-mutex-stress	pr/C22-mutex-stress-ram	c8fbb351	test	S4 test	test(pico2): mutex-stress
25	pr/C22-rtos-apis-ram	pr/C22-mutex-stress	0a2e9b77	test	S4 test	test(pico2): rtos-apis-ram
26	pr/C22-sc-test-ko	pr/C22-rtos-apis-ram	89a5454d	test	S4 test	test(pico2): sc-test-ko
27	pr/C22-smp-mat-test	pr/C22-sc-test-ko	57149d31	test	S4 test	test(pico2): smp-mat-test
28	pr/C22-smp-mat-test-ram	pr/C22-smp-mat-test	d3bd1bff	test	S4 test	test(pico2): smp-mat-test-ram
29	pr/C22-smp-test-ko	pr/C22-smp-mat-test-ram	d4fb4e67	test	S4 test	test(pico2): smp-test-ko
30	pr/C22-smp-test-usb-cdc-acm	pr/C22-smp-test-ko	6156aa38	test	S4 test	test(pico2): smp-test-usb-cdc-acm
31	pr/C22-smp-test-usb-hid	pr/C22-smp-test-usb-cdc-acm	385cb24c	test	S4 test	test(pico2): smp-test-usb-hid
32	pr/C22-smp-test0	pr/C22-smp-test-usb-hid	e7546968	test	S4 test	test(pico2): smp-test0
33	pr/C22-smp-test1	pr/C22-smp-test0	a8263f0d	test	S4 test	test(pico2): smp-test1

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
34	pr/C22-smp-test2	pr/C22-smp-test1	137a1cb1	test	S4 test	test(pico2): smp-test2
35	pr/C22-smp-test3	pr/C22-smp-test2	802c05fd	test	S4 test	test(pico2): smp-test3
36	pr/C22-smp-test4	pr/C22-smp-test3	30d31e49	test	S4 test	test(pico2): smp-test4
37	pr/C22-smp-test5	pr/C22-smp-test4	9367c2fa	test	S4 test	test(pico2): smp-test5
38	pr/C23	pr/C22-smp-test5	1cffd384	test	S4 board/ harness	test(pico2-rp2350b-psram): the RP2350B board with 16 MiB flash and PSRAM
39	pr/C23-exc-test	pr/C23	ecbfb0b0	test	S4 test	test(pico2-rp2350b-psram): exc-test
40	pr/C23-fp-switch	pr/C23-exc-test	f2708769	test	S4 test	test(pico2-rp2350b-psram): fp-switch
41	pr/C23-sc-test-ko	pr/C23-fp-switch	9767e0dc	test	S4 test	test(pico2-rp2350b-psram): sc-test-ko
42	pr/C23-smp-mat-test	pr/C23-sc-test-ko	5a78743c	test	S4 test	test(pico2-rp2350b-psram): smp-mat-test
43	pr/C23-smp-test-ko	pr/C23-smp-mat-test	eb888556	test	S4 test	test(pico2-rp2350b-psram): smp-test-ko
44	pr/C23-smp-test-nested-clock	pr/C23-smp-test-ko	1c50c6c1	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested-clock
45	pr/C23-smp-test-nested-clock_200	pr/C23-smp-test-nested-clock	dcc8fdf3	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested-clock_200
46			53e7990b	test	S4 test	test(pico2-rp2350b-

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
	<code>pr/C23-smp-test-nested-clock_250</code>	<code>pr/C23-smp-test-nested-clock_200</code>				<code>psram): smp-test-nested-clock_250</code>
47	<code>pr/C23-smp-test-nested</code>	<code>pr/C23-smp-test-nested-clock_250</code>	<code>03462dd0</code>	test	S4 test	<code>test(pico2-rp2350b-psram): smp-test-nested</code>
48	<code>pr/C23-smp-test0</code>	<code>pr/C23-smp-test-nested</code>	<code>4fddd1a6</code>	test	S4 test	<code>test(pico2-rp2350b-psram): smp-test0</code>
49	<code>pr/C23-smp-test1</code>	<code>pr/C23-smp-test0</code>	<code>38f862a7</code>	test	S4 test	<code>test(pico2-rp2350b-psram): smp-test1</code>
50	<code>pr/C23-smp-test2</code>	<code>pr/C23-smp-test1</code>	<code>792ee80b</code>	test	S4 test	<code>test(pico2-rp2350b-psram): smp-test2</code>
51	<code>pr/C23-smp-test3</code>	<code>pr/C23-smp-test2</code>	<code>fdcd879c</code>	test	S4 test	<code>test(pico2-rp2350b-psram): smp-test3</code>
52	<code>pr/C23-smp-test4</code>	<code>pr/C23-smp-test3</code>	<code>0b2d5838</code>	test	S4 test	<code>test(pico2-rp2350b-psram): smp-test4</code>
53	<code>pr/C23-smp-test5</code>	<code>pr/C23-smp-test4</code>	<code>15867310</code>	test	S4 test	<code>test(pico2-rp2350b-psram): smp-test5</code>
54	<code>pr/C24</code>	<code>pr/C23-smp-test5</code>	<code>994cb864</code>	test	S4 board/ harness	<code>test(pico2-pizero): the RP2350 on a Pi Zero form-factor carrier</code>
55	<code>pr/C24-exc-test</code>	<code>pr/C24</code>	<code>e450aaff</code>	test	S4 test	<code>test(pico2-pizero): exc-test</code>
56	<code>pr/C24-psram-exec</code>	<code>pr/C24-exc-test</code>	<code>ab90bf91</code>	test	S4 test	<code>test(pico2-pizero): psram-exec</code>
57	<code>pr/C24-psram-mat-test-250</code>	<code>pr/C24-psram-exec</code>	<code>76a2d22f</code>	test	S4 test	<code>test(pico2-pizero): psram-mat-test-250</code>

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
58	<code>pr/C24-sc-test-ko</code>	<code>pr/C24-psram-mat-test-250</code>	<code>25cb812e</code>	test	S4 test	<code>test(pico2-pizero): sc-test-ko</code>
59	<code>pr/C24-smp-mat-test</code>	<code>pr/C24-sc-test-ko</code>	<code>43d09f78</code>	test	S4 test	<code>test(pico2-pizero): smp-mat-test</code>
60	<code>pr/C24-smp-test-ko</code>	<code>pr/C24-smp-mat-test</code>	<code>395b0d10</code>	test	S4 test	<code>test(pico2-pizero): smp-test-ko</code>
61	<code>pr/C24-smp-test-usb-cdc-acm</code>	<code>pr/C24-smp-test-ko</code>	<code>038509f6</code>	test	S4 test	<code>test(pico2-pizero): smp-test-usb-cdc-acm</code>
62	<code>pr/C24-smp-test-usb-hid</code>	<code>pr/C24-smp-test-usb-cdc-acm</code>	<code>1eea852d</code>	test	S4 test	<code>test(pico2-pizero): smp-test-usb-hid</code>
63	<code>pr/C24-smp-test0</code>	<code>pr/C24-smp-test-usb-hid</code>	<code>044c5bde</code>	test	S4 test	<code>test(pico2-pizero): smp-test0</code>
64	<code>pr/C24-smp-test1</code>	<code>pr/C24-smp-test0</code>	<code>12570a64</code>	test	S4 test	<code>test(pico2-pizero): smp-test1</code>
65	<code>pr/C24-smp-test2</code>	<code>pr/C24-smp-test1</code>	<code>924b36a0</code>	test	S4 test	<code>test(pico2-pizero): smp-test2</code>
66	<code>pr/C24-smp-test3</code>	<code>pr/C24-smp-test2</code>	<code>807afed1</code>	test	S4 test	<code>test(pico2-pizero): smp-test3</code>
67	<code>pr/C24-smp-test4</code>	<code>pr/C24-smp-test3</code>	<code>cb72fb3e</code>	test	S4 test	<code>test(pico2-pizero): smp-test4</code>
68	<code>pr/C24-smp-test5</code>	<code>pr/C24-smp-test4</code>	<code>b6459bcb</code>	test	S4 test	<code>test(pico2-pizero): smp-test5</code>
69	<code>pr/C25-cmsis-os-validator</code>	<code>pr/C24-smp-test5</code>	<code>14bba6b7</code>	test	S4 test	<code>test(nucleof411): cmsis-os-validator</code>
70	<code>pr/C25-mos-test1</code>	<code>pr/C25-cmsis-os-validator</code>	<code>82c09863</code>	test	S4 test	<code>test(nucleof411): mos-test1</code>
71	<code>pr/C25-mutex-stress</code>	<code>pr/C25-mos-test1</code>	<code>5a3c8ec6</code>	test	S4 test	<code>test(nucleof411): mutex-stress</code>

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
72	pr/C25-rtos-apis	pr/C25-mutex-stress	904bc935	test	S4 test	test(nucleof411): rtos-apis
73	pr/C26	pr/C25-rtos-apis	b695919a	test	S4 board/ harness	test(weactf411): the WeAct Black Pill F411 board
74	pr/C26-cmsis-os-validator	pr/C26	3edc06c0	test	S4 test	test(weactf411): cmsis-os-validator
75	pr/C26-mos-test1	pr/C26-cmsis-os-validator	900c9402	test	S4 test	test(weactf411): mos-test1
76	pr/C26-mutex-stress	pr/C26-mos-test1	7b9526bb	test	S4 test	test(weactf411): mutex-stress
77	pr/C26-rtos-apis	pr/C26-mutex-stress	c8d67d58	test	S4 test	test(weactf411): rtos-apis
78	pr/C26-spi-pipeline	pr/C26-rtos-apis	6935c185	test	S4 test	test(weactf411): spi-pipeline
79	pr/C27	pr/C26-spi-pipeline	dc22fcdb	test	S4 board/ harness	test(weactf412): the WeAct F412 board
80	pr/C27-cmsis-os-validator	pr/C27	a5f805b5	test	S4 test	test(weactf412): cmsis-os-validator
81	pr/C27-mos-test1	pr/C27-cmsis-os-validator	f2f0695c	test	S4 test	test(weactf412): mos-test1
82	pr/C27-mutex-stress	pr/C27-mos-test1	efa2b9a7	test	S4 test	test(weactf412): mutex-stress
83	pr/C27-rtos-apis	pr/C27-mutex-stress	bc32b726	test	S4 test	test(weactf412): rtos-apis
84	pr/C27-uart-test1	pr/C27-rtos-apis	efd8432c	test	S4 test	test(weactf412): uart-test1
85	pr/C28	pr/C27-uart-test1	72149579	chore	S5 docs/chore	chore: ignore every build*

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						directory, the build products and the USB host tool
86	pr/C29	pr/C28	c5a71fbc	docs	S5 docs/chore	docs: README for the six boards and the generic QEMU cores

micro-os-plus-iii-aarch32 -- 64 pull requests (github.com/dan-maio/micro-os-plus-iii-aarch32)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/A32-01	xpack-development	03a0cfc1	new code	S0 devices	<code>feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers</code>
2	pr/A32-02	pr/A32-01	c318b741	new code	S0 devices	<code>feat(soc): import the BCM2837 (Raspberry Pi 3 B / Zero 2 W) SoC support</code>
3	pr/A32-03	pr/A32-02	58cff47a	new code	S0 devices	<code>feat(soc): import the RK3506 (Luckfox Lyra) SoC support</code>
4	pr/A32-04	pr/A32-03	a8662d6e	chore	S3 port	<code>chore: ignore every build* directory</code>
5	pr/A32-05	pr/A32-04	b0eb0ed7	new code	S3 port	<code>feat(port): AArch32 port declarations ' application config and device header</code>
6	pr/A32-06	pr/A32-05	81d97c4d	new code	S3 port	<code>feat(port): ARMv7-A MMU bring-up</code>
7	pr/A32-07	pr/A32-06	583aa440	new code	S3 port	<code>feat(port): AArch32 exception vectors, IRQ dispatch and fault dump</code>
8	pr/A32-08	pr/A32-07	4d9e3c1b	new code	S3 port	<code>feat(port): ARM generic timer, and the high-resolution counter from CNTPCT</code>
9	pr/A32-09	pr/A32-08	e510bd9f	new code	S3 port	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						feat(port): AArch32 context switch and the port half of the SMP scheduler
10	pr/A32-10	pr/A32-09	37b944cf	new code	S3 port	feat(port): secondary- core entry
11	pr/A32-11	pr/A32-10	6eb8c2aa	new code	S3 port	feat(port): semihosting console and a strong _Exit()
12	pr/A32-20	pr/A32-11	8cac0c56	test	S4 board/ harness	test(rpi- zero-2w): the Raspberry Pi Zero 2 W board (BCM2837, 4x Cortex-A53, AArch32)
13	pr/A32-21	pr/A32-20	576597f0	build	S4 board/ harness	build(cmake) : the AArch32 port project, its board model and its test builder
14	pr/A32-T- smp_test0	pr/A32-21	0308bd45	test	S4 test	test(rpi- zero-2w): smp_test0
15	pr/A32-T- smp-mat- test	pr/A32-T- smp_test0	3f8b9a64	test	S4 test	test(rpi- zero-2w): smp-mat- test
16	pr/A32-T- cmsis-os- validator	pr/A32-T- smp-mat- test	8bccb4ee	test	S4 test	test(rpi- zero-2w): cmsis-os- validator
17	pr/A32-T- mutex- stress	pr/A32-T- cmsis-os- validator	cc8d43bb	test	S4 test	test(rpi- zero-2w): mutex- stress
18	pr/A32-T- rtos-apis		f53d5e9e	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
		pr/A32-T-mutex-stress				test(rpi-zero-2w): rtos-apis
19	pr/A32-T-sd_test	pr/A32-T-rtos-apis	286da73d	test	S4 test	test(rpi-zero-2w): sd_test
20	pr/A32-T-smp-mat-sdcard-test	pr/A32-T-sd_test	2bd8db32	test	S4 test	test(rpi-zero-2w): smp-mat-sdcard-test
21	pr/A32-T-smp-num-test	pr/A32-T-smp-mat-sdcard-test	4127ec3c	test	S4 test	test(rpi-zero-2w): smp-num-test
22	pr/A32-T-smp-pipeline-test	pr/A32-T-smp-num-test	89a1b78c	test	S4 test	test(rpi-zero-2w): smp-pipeline-test
23	pr/A32-T-smp-pro-cons-test	pr/A32-T-smp-pipeline-test	87ca7814	test	S4 test	test(rpi-zero-2w): smp-pro-cons-test
24	pr/A32-T-smp_test1	pr/A32-T-smp-pro-cons-test	98efc7ea	test	S4 test	test(rpi-zero-2w): smp_test1
25	pr/A32-T-smp_test2	pr/A32-T-smp_test1	3f4148c8	test	S4 test	test(rpi-zero-2w): smp_test2
26	pr/A32-T-smp_test3	pr/A32-T-smp_test2	d4443aca	test	S4 test	test(rpi-zero-2w): smp_test3
27	pr/A32-T-smp_test4	pr/A32-T-smp_test3	bd38979a	test	S4 test	test(rpi-zero-2w): smp_test4
28	pr/A32-T-usb_test	pr/A32-T-smp_test4	aa7fae53	test	S4 test	test(rpi-zero-2w): usb_test
29	pr/A32-R3	pr/A32-T-usb_test	aa0b83c0	test	S4 board/ harness	test(rpi3b): the Raspberry Pi 3 B board (AArch32), sharing the Zero 2 W sources
30		pr/A32-R3	22010a2c	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
	pr/A32-R3-smp_test0					test(rpi3b): smp_test0
31	pr/A32-R3-cmsis-os-validator	pr/A32-R3-smp_test0	e0df39c7	test	S4 test	test(rpi3b): cmsis-os-validator
32	pr/A32-R3-mutex-stress	pr/A32-R3-cmsis-os-validator	b3666f39	test	S4 test	test(rpi3b): mutex-stress
33	pr/A32-R3-rtos-apis	pr/A32-R3-mutex-stress	832c71ae	test	S4 test	test(rpi3b): rtos-apis
34	pr/A32-R3-sd_test	pr/A32-R3-rtos-apis	8de31552	test	S4 test	test(rpi3b): sd_test
35	pr/A32-R3-smp-mat-sdcard-test	pr/A32-R3-sd_test	465e7d4a	test	S4 test	test(rpi3b): smp-mat-sdcard-test
36	pr/A32-R3-smp-mat-test	pr/A32-R3-smp-mat-sdcard-test	c80191d9	test	S4 test	test(rpi3b): smp-mat-test
37	pr/A32-R3-smp-num-test	pr/A32-R3-smp-mat-test	4359e4ce	test	S4 test	test(rpi3b): smp-num-test
38	pr/A32-R3-smp-pipeline-test	pr/A32-R3-smp-num-test	385f0938	test	S4 test	test(rpi3b): smp-pipeline-test
39	pr/A32-R3-smp-pro-cons-test	pr/A32-R3-smp-pipeline-test	b5235187	test	S4 test	test(rpi3b): smp-pro-cons-test
40	pr/A32-R3-smp_test1	pr/A32-R3-smp-pro-cons-test	436d38c9	test	S4 test	test(rpi3b): smp_test1
41	pr/A32-R3-smp_test2	pr/A32-R3-smp_test1	a121a4ca	test	S4 test	test(rpi3b): smp_test2
42	pr/A32-R3-smp_test3	pr/A32-R3-smp_test2	6b56b17f	test	S4 test	test(rpi3b): smp_test3
43	pr/A32-R3-smp_test4	pr/A32-R3-smp_test3	bd4569eb	test	S4 test	test(rpi3b): smp_test4
44	pr/A32-R3-usb_test	pr/A32-R3-smp_test4	a33a5cc0	test	S4 test	test(rpi3b): usb_test
45	pr/A32-LL	pr/A32-R3-usb_test	b468cbba	test	S4 board/ harness	test(luckfox-lyra): the Luckfox Lyra B board (RK3506, 3x

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						Cortex-A7, hardware only)
46	pr/A32-LL- smp_test0	pr/A32-LL	e617e017	test	S4 test	test(luckfox -lyra): smp_test0
47	pr/A32-LL- smp_test_int 5	pr/A32-LL- smp_test0	067f4a19	test	S4 test	test(luckfox -lyra): smp_test_int 5
48	pr/A32-LL- sd_test	pr/A32-LL- smp_test_int 5	79eca794	test	S4 test	test(luckfox -lyra): sd_test
49	pr/A32-LL- smp-mat- sdcard-test	pr/A32-LL- sd_test	aa5a77ac	test	S4 test	test(luckfox -lyra): smp- mat-sdcard- test
50	pr/A32-LL- smp-mat- test	pr/A32-LL- smp-mat- sdcard-test	fcaeda34	test	S4 test	test(luckfox -lyra): smp- mat-test
51	pr/A32-LL- smp-num- test	pr/A32-LL- smp-mat- test	3f2f7a29	test	S4 test	test(luckfox -lyra): smp- num-test
52	pr/A32-LL- smp- pipeline- test	pr/A32-LL- smp-num- test	7c9a0205	test	S4 test	test(luckfox -lyra): smp- pipeline- test
53	pr/A32-LL- smp-pro- cons-test	pr/A32-LL- smp- pipeline- test	b42eaf86	test	S4 test	test(luckfox -lyra): smp- pro-cons- test
54	pr/A32-LL- smp_test1	pr/A32-LL- smp-pro- cons-test	bd934140	test	S4 test	test(luckfox -lyra): smp_test1
55	pr/A32-LL- smp_test2	pr/A32-LL- smp_test1	09b7ce26	test	S4 test	test(luckfox -lyra): smp_test2
56	pr/A32-LL- smp_test3	pr/A32-LL- smp_test2	0b58a238	test	S4 test	test(luckfox -lyra): smp_test3
57	pr/A32-LL- smp_test4	pr/A32-LL- smp_test3	9a0a34a0	test	S4 test	test(luckfox -lyra): smp_test4
58	pr/A32-LL- smp_test5	pr/A32-LL- smp_test4	979865e4	test	S4 test	test(luckfox -lyra): smp_test5

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
59	pr/A32-LL-smp_test6	pr/A32-LL-smp_test5	1cdf9eb1	test	S4 test	test(luckfox-lyra): smp_test6
60	pr/A32-LL-smp_test7	pr/A32-LL-smp_test6	0a1f7b1e	test	S4 test	test(luckfox-lyra): smp_test7
61	pr/A32-LL-smp_test_int	pr/A32-LL-smp_test7	3ce4a94a	test	S4 test	test(luckfox-lyra): smp_test_int
62	pr/A32-LL-smp_test_int2	pr/A32-LL-smp_test_int	d3f74a0f	test	S4 test	test(luckfox-lyra): smp_test_int2
63	pr/A32-LL-smp_test_int3	pr/A32-LL-smp_test_int2	d254ad93	test	S4 test	test(luckfox-lyra): smp_test_int3
64	pr/A32-LL-smp_test_int4	pr/A32-LL-smp_test_int3	0aa86aef	test	S4 test	test(luckfox-lyra): smp_test_int4

micro-os-plus-iii-aarch64 -- 43 pull requests (github.com/dan-maio/micro-os-plus-iii-aarch64)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/A64-01	xpack-development	b764bc98	new code	S0 devices	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers
2	pr/A64-02	pr/A64-01	fe04ec15	new code	S0 devices	feat(soc): import the BCM2837 (Raspberry Pi 3 B / Zero 2 W) SoC support
3	pr/A64-03	pr/A64-02	30822be2	chore	S3 port	chore: ignore every build* directory
4	pr/A64-04	pr/A64-03	f04d5ef5	new code	S3 port	feat(port): AArch64 port declarations , application config and device header
5	pr/A64-05	pr/A64-04	522cf173	new code	S3 port	feat(port): AArch64 MMU bring-up
6	pr/A64-06	pr/A64-05	265ce322	new code	S3 port	feat(port): AArch64 exception reporting and the static constructor runners
7	pr/A64-07	pr/A64-06	f5359a25	new code	S3 port	feat(port): ARM generic timer accessors (CNTFRQ_EL0, CNTPCT_EL0)
8	pr/A64-08	pr/A64-07	e04efa95	new code	S3 port	feat(port): AArch64 context frame and the port half of the

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						SMP scheduler
9	pr/A64-09	pr/A64-08	ba24acc6	new code	S3 port	feat(port): secondary-core entry
10	pr/A64-10	pr/A64-09	7e796801	new code	S3 port	feat(port): semihosting console
11	pr/A64-20	pr/A64-10	9cafe2de	test	S4 board/harness	test(rpi-zero-2w): the Raspberry Pi Zero 2 W board (BCM2837, 4x Cortex-A53, AArch64)
12	pr/A64-21	pr/A64-20	475d0604	build	S4 board/harness	build(cmake): the AArch64 port project, its board model and its test builder
13	pr/A64-T-smp_test0	pr/A64-21	cca66872	test	S4 test	test(rpi-zero-2w): smp_test0
14	pr/A64-T-smp-mat-test	pr/A64-T-smp_test0	07d27cd6	test	S4 test	test(rpi-zero-2w): smp-mat-test
15	pr/A64-T-cmsis-os-validator	pr/A64-T-smp-mat-test	bc32bcd8	test	S4 test	test(rpi-zero-2w): cmsis-os-validator
16	pr/A64-T-mutex-stress	pr/A64-T-cmsis-os-validator	d87cebb7	test	S4 test	test(rpi-zero-2w): mutex-stress
17	pr/A64-T-rtos-apis	pr/A64-T-mutex-stress	99c5a5bf	test	S4 test	test(rpi-zero-2w): rtos-apis
18	pr/A64-T-sd_test	pr/A64-T-rtos-apis	d666d390	test	S4 test	test(rpi-zero-2w): sd_test
19	pr/A64-T-smp-mat-sdcard-test	pr/A64-T-sd_test	1731d0b1	test	S4 test	test(rpi-zero-2w):

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						smp-mat- sdcard-test
20	pr/A64-T- smp-num- test	pr/A64-T- smp-mat- sdcard-test	edf06551	test	S4 test	test(rpi- zero-2w): smp-num- test
21	pr/A64-T- smp- pipeline- test	pr/A64-T- smp-num- test	8750df6e	test	S4 test	test(rpi- zero-2w): smp- pipeline- test
22	pr/A64-T- smp-pro- cons-test	pr/A64-T- smp- pipeline- test	5e0e5a65	test	S4 test	test(rpi- zero-2w): smp-pro- cons-test
23	pr/A64-T- smp_test1	pr/A64-T- smp-pro- cons-test	ed02a26b	test	S4 test	test(rpi- zero-2w): smp_test1
24	pr/A64-T- smp_test2	pr/A64-T- smp_test1	12c23d59	test	S4 test	test(rpi- zero-2w): smp_test2
25	pr/A64-T- smp_test3	pr/A64-T- smp_test2	216918dd	test	S4 test	test(rpi- zero-2w): smp_test3
26	pr/A64-T- smp_test4	pr/A64-T- smp_test3	87d619e3	test	S4 test	test(rpi- zero-2w): smp_test4
27	pr/A64-T- usb_test	pr/A64-T- smp_test4	5da24f9a	test	S4 test	test(rpi- zero-2w): usb_test
28	pr/A64-R3	pr/A64-T- usb_test	c84b8e00	test	S4 board/ harness	test(rpi3b): the Raspberry Pi 3 B board (AArch64), sharing the Zero 2 W sources
29	pr/A64-R3- smp_test0	pr/A64-R3	b3fc5a12	test	S4 test	test(rpi3b): smp_test0
30	pr/A64-R3- cmsis-os- validator	pr/A64-R3- smp_test0	031d8f52	test	S4 test	test(rpi3b): cmsis-os- validator
31	pr/A64-R3- mutex- stress	pr/A64-R3- cmsis-os- validator	3ae3d571	test	S4 test	test(rpi3b): mutex- stress
32			e7bc6bc7	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
	pr/A64-R3-rtos-apis	pr/A64-R3-mutex-stress				test(rpi3b):rtos-apis
33	pr/A64-R3-sd_test	pr/A64-R3-rtos-apis	3e9f6e5a	test	S4 test	test(rpi3b):sd_test
34	pr/A64-R3-smp-mat-sdcard-test	pr/A64-R3-sd_test	ea725894	test	S4 test	test(rpi3b):smp-mat-sdcard-test
35	pr/A64-R3-smp-mat-test	pr/A64-R3-smp-mat-sdcard-test	0cce173e	test	S4 test	test(rpi3b):smp-mat-test
36	pr/A64-R3-smp-num-test	pr/A64-R3-smp-mat-test	d4f629c3	test	S4 test	test(rpi3b):smp-num-test
37	pr/A64-R3-smp-pipeline-test	pr/A64-R3-smp-num-test	a8082ca9	test	S4 test	test(rpi3b):smp-pipeline-test
38	pr/A64-R3-smp-pro-cons-test	pr/A64-R3-smp-pipeline-test	1ace669b	test	S4 test	test(rpi3b):smp-pro-cons-test
39	pr/A64-R3-smp_test1	pr/A64-R3-smp-pro-cons-test	84bceba9	test	S4 test	test(rpi3b):smp_test1
40	pr/A64-R3-smp_test2	pr/A64-R3-smp_test1	c045d19b	test	S4 test	test(rpi3b):smp_test2
41	pr/A64-R3-smp_test3	pr/A64-R3-smp_test2	fe5a5489	test	S4 test	test(rpi3b):smp_test3
42	pr/A64-R3-smp_test4	pr/A64-R3-smp_test3	ebf7dce3	test	S4 test	test(rpi3b):smp_test4
43	pr/A64-R3-usb_test	pr/A64-R3-smp_test4	a723ecc1	test	S4 test	test(rpi3b):usb_test

micro-os-plus-iii-devices -- 1 pull request (github.com/dan-maio/micro-os-plus-iii-devices)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/D01	smp	(bundle)	docs	S0 devices	docs: retire the repository -- its contents moved to the architecture ports

D01 is kept in `devices-retired.bundle` (the local clone was deleted after S0, §14.2). After it is merged, the repository is archived on GitHub.

22.4 Documentation updates after the migration (kernel PRs 126–159)

Made after the full test run, so that every document in `docs/` describes the executed state: Part III itself (K985); one commit per document (K986-), *which corrects stale facts in the reference documents (no `micro-os-plus-iii-devices`, no `UOS_DEVICES_DIR`, the current clone layout and the verified results) and adds a Status 2026-10-06* note to the historical plans and reviews (their PDFs regenerated)*; the two commits that complete this section (K987, K988); and the verification record of §21.6 (K989). The upstream documents that do not concern the migration (`HISTORY`, `NOTES`, `TODO`, `posix-io-*`, `other-posix-systems`, `upstream-*`) are unchanged.

23. Regeneration of 2026-10-07: from the clone to the last `pr/` commit

The `smp` branches of all repositories were regenerated once more from `xpack-development`, in an empty `/tmp`. This section records how, the branches that resulted, and the test run.

23.1 Summary of the generation

Step	What was done	Result
1	Clone (§12.1): the six repositories from <code>github.com/dan-maio</code> ; in the five code repositories the read-only <code>golden</code> remote (<code>~/Work2/micro-os-plus/<repo></code>) and <code>git fetch golden smp</code> .	rc 0
2	Branch <code>smp</code> from <code>origin/xpack-development</code> . In <code>micro-os-plus-iii</code> and <code>micro-os-plus-iii-posix-arch</code> the GitHub default branch is <code>smp</code> , so the clone already has a local <code>smp</code> ; there <code>git switch -C smp origin/xpack-development</code> replaces <code>git switch -c</code> .	bases as §20.1 (table 23.2)
3	aarch64 base check: <code>git log --oneline 366c957...fe904c0</code> gives one commit, <code>fe904c0 package.json update</code> ; <code>fe904c0</code> is used, as in §20.1.	—
4	Install (§13.3): <code>xpm install && xpm link</code> in <code>posix-arch</code> and <code>cortexm</code> , <code>xpm run install-all</code> in <code>micro-os-plus-iii/tests</code> ; <code>aarch32</code> and <code>aarch64</code> <code>xpm install && xpm link</code> before the test run.	rc 0 for all
5	Commits: the commits of §22.3 were applied in §22.3 order with <code>git cherry-pick</code> , from the recorded branches of the 2026-10-06 replay (§21.6). Each new commit was compared with its recorded commit: the tree is identical at every commit, and Commit-ID, subject and order match §22.3 (376 rows; K989's SHA column reads "(this commit)").	376 commits, 0 tree differences
6	Branches: one local branch <code>pr/<Commit-ID></code> per commit (table 23.3).	376 branches
7	Devices (§14.2): D01 applied to the fresh devices clone, branch <code>pr/D01</code> , kept in <code>devices-retired-2026-10-07.bundle</code> ; then the local devices clone was deleted.	—
8	Full test run (latest compilers, <code>ctest -LE hwd</code>): table 23.4.	all green

Only the SHAs differ from §22.3, because the commit dates differ. Nothing was pushed.

23.2 Bases and results

Repository	Base (<code>origin/xpack-development</code>)	Commits on <code>smp</code>	First PR branch	Last PR branch (<code>smp</code> tip)
<code>micro-os-plus-iii</code>	<code>7f1ce5ca</code>	159	<code>pr/K01</code> <code>deb7dba1</code>	<code>pr/K989</code> <code>ccf87e89</code>
<code>micro-os-plus-iii-posix-arch</code>	<code>86a6a1f</code>	24	<code>pr/P01</code> <code>a2455e0a</code>	<code>pr/P36</code> <code>4ed05ad9</code>
<code>micro-os-plus-iii-cortexm</code>	<code>687e975</code>	86	<code>pr/C01</code> <code>995592d4</code>	<code>pr/C29</code> <code>71e98ff7</code>
<code>micro-os-plus-iii-aarch32</code>	<code>307c62f</code>	64	<code>pr/A32-01</code> <code>fba3b274</code>	<code>pr/A32-LL-smp_test_int4</code> <code>e721deab</code>
<code>micro-os-plus-iii-aarch64</code>	<code>fe904c0</code>	43	<code>pr/A64-01</code> <code>fd554718</code>	<code>pr/A64-R3-usb_test</code> <code>8151c26a</code>
<code>micro-os-plus-iii-devices</code>	<code>e8e39d5</code> (<code>smp</code>)	1	<code>pr/D01</code> (bundle)	<code>pr/D01</code> (bundle)

This section is one more kernel commit after K989, `K990` (`pr/K990`), so the kernel `smp` tip is K990.

23.3 The `pr/` branches

- **One branch per commit.** Every commit of `smp` has a local branch `pr/<Commit-ID>`, where `<Commit-ID>` is the ID in the commit trailer `Commit-ID: <ID>` (`xpack-dev-smp.md Part II`). Checked in each repository: number of `pr/*` branches = number of commits, no commit without a branch, no two branches on the same commit.
- **Stacked.** `pr/<ID>` points at its commit, whose parent is the previous commit, i.e. the previous `pr/` branch. The first branch of a repository sits on `xpack-development`; the last one is the `smp` tip.
- **One branch = one PR.** The PR for `pr/<ID>` has the previous `pr/` branch as its base (the first one has `xpack-development`); after a merge, the next PR is retargeted (§12.5). The cross-repository merge order is §22.2.
- **Local only.** The branches exist in each `/tmp/<repository>` clone; they are pushed to `github.com/dan-maio/<repository>` only when the PRs are opened.
- **This run's SHAs.** §22.3 lists the SHAs of the 2026-10-06 replay. The SHAs of this run are listed in each clone with:

```
git for-each-ref --sort=committerdate --format='%(%(refname:short) %(objectname:short=8) %(subject))' 'refs/heads/pr/*'
```

23.4 Full test run (μOS++ test framework, `ctest -LE hwd`)

Compilers: host gcc 16.2.1 for `native-cmake-sys`; xPack `arm-none-eabi-gcc` and `aarch64-none-elf-gcc` 15.2.1 for the cross builds.

Action (in <code>micro-os-plus-iii/tests</code>)	Debug	Release
<code>xpm run test-native-cmake-sys</code>	16/16	16/16
<code>xpm run test-cortex-cmake</code> (qemu-cortex-m0, m3, m4f, m7f)	3/3 each	3/3 each
<code>test-aarch32-rpi-zero-2w-cmake</code>	15/15	15/15
<code>test-aarch32-rpi3b-cmake</code>	15/15	15/15
<code>test-aarch64-rpi-zero-2w-cmake</code>	15/15	15/15
<code>test-aarch64-rpi3b-cmake</code>	15/15	15/15 (see below)
<code>test-2xcortex-m33-cmake</code>	4/4	4/4
<code>test-pico2-1cpu-cmake</code>	4/4	4/4
<code>test-cortexm-pico2-cmake</code>	6/6	6/6
<code>test-cortexm-pico2-rp2350b-psram-cmake</code>	3/3	3/3

In the `xpm run test-smp-cmake` pass, `aarch64-rpi3b-smp-pipeline-test-qemu` (release) stopped after 5.5 s: QEMU exited with rc 1 after the `t= 4347 ms` progress line, without a `RESULT` line, and the action stopped there. Re-run alone, unchanged, it passed (35.6 s). The four remaining platforms were then run with their own `xpm run test-<platform>-cmake` actions, all rc 0. Like the stall recorded in §21.6, this is an intermittent behaviour of the SMP pipeline test under QEMU, to be investigated separately.